



Additional Problems for Self Practice (APSP)

✎ Marked questions are recommended for Revision.

This Section is not meant for classroom discussion. It is being given to promote self-study and self testing amongst the Resonance students.

PART - I : PRACTICE TEST-1 (IIT-JEE (MAIN Pattern))

Max. Time : 1 Hr.

Max. Marks : 120

Important Instructions

- The test is of **1 hour** duration.
- The Test Booklet consists of **30** questions. The maximum marks are **120**.
- Each question is allotted **4 (four)** marks for correct response.
- Candidates will be awarded marks as stated above in Instructions No. 3 for correct response of each question. $\frac{3}{4}$ (**one fourth**) marks will be deducted for indicating incorrect response of each question. No deduction from the total score will be made if no response is indicated for an item in the answer sheet.
- There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 4 above.

- If the electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$, then the electrons involved in bonding will be.
 (1) $3p^6$ (2) $3p^6 4s^2$ (3) $3p^6 3d^2$ (4) $3d^2 4s^2$
- Two ice cubes are pressed together until they form one block. Which of the following force is primarily responsible for holding the cubes together?
 (1) Dipole-dipole interaction (2) Van der Waals forces
 (3) Ionic interaction (4) Hydrogen bonding
- ✎ A sigma bond may be formed by the overlap of 2 atomic orbitals of atoms A and B. If the bond is formed along the x-axis, which of the following overlaps is acceptable ?
 (1) s orbital of A and p_z orbital of B (2) p_x orbital of A and p_y orbital of B
 (3) p_z orbital of A and p_x orbital of B (4) p_x orbital of A and s orbital of B
- NH_3 and BF_3 combine readily because of the formation of :
 (1) π -back bond (2) a hydrogen bond
 (3) a coordinate bond (4) an ionic bond
- Maximum bond energy is in :
 (1) F_2 (2) N_2 (3) O_2 (4) equal in all
- The structure of XeF_2 involves hybridization of the type :
 (1) sp^3 (2) sp^3d^3 (3) sp^3d (4) sp^3d^2
- ✎ The bond angle and hybridization in ether (CH_3OCH_3) is :
 (1) $106^\circ 51'$, sp^3 (2) $104^\circ 31'$, sp^3 (3) $109^\circ 28'$, sp^3 (4) $>109^\circ 28'$, sp^3
- Which of the following have maximum boiling point
 (1) CH_4 (2) CF_4 (3) CCl_4 (4) CD_4
- In which of the following molecule / ion all the bonds are not equal ?
 (1) XeF_4 (2) BeF_4^- (3) C_2H_4 (4) SiF_4
- ✎ The correct order of increasing X–O–X bond angle is (X = H, F or Cl) :
 (1) $H_2O > Cl_2O > F_2O$ (2) $Cl_2O > H_2O > F_2O$
 (3) $F_2O > Cl_2O > H_2O$ (4) $F_2O > H_2O > Cl_2O$
- Which of the following is paramagnetic ?
 (1) O_2^- (2) CN^- (3) CO (4) NO^+





12. Which of the following is non-polar :
 (1) NF_3 (2) BF_3 (3) PF_3 (4) SF_4
13. Strongest hydrogen bond present in :
 (1) $\text{O}-\text{H} \cdots \text{S}$ (2) $\text{S}-\text{H} \cdots \text{O}$ (3) $\text{F}-\text{H} \cdots \text{F}$ (4) $\text{F}-\text{H} \cdots \text{O}$
14. Resonance structure of a molecule should not have
 (1) Identical bond position (2) Identical arrangements of atoms
 (3) Nearly the same energy content (4) The same number of paired electrons
15. The paramagnetic property of oxygen is well explained by :
 (1) Molecular orbital theory (2) Resonance theory
 (3) Valence bond theory (4) VSEPR theory
16. Which of the following statement is correct regarding molecular orbital theory (MOT) :
 (1) Energy of bonding orbital is less than anti-bonding orbital.
 (2) Energy of bonding orbital is more than anti-bonding orbital.
 (3) Bonding orbitals are monocentric.
 (4) Bonding orbital follow $n + \ell$ rule
17. PF_2Cl_3 is non polar because :
 (1) P-Cl bond is non-polar (2) Its dipole moment is zero
 (3) P-Cl bond is polar (4) P & Cl have equal electronegativity
18. The hybrid states of central atom in diborane, diamond and graphite are respectively :
 (1) $\text{sp}^2, \text{sp}^3, \text{sp}^2$ (2) $\text{sp}^3, \text{sp}^3, \text{sp}^2$ (3) $\text{sp}^3, \text{sp}^3, \text{sp}^3$ (4) $\text{sp}, \text{sp}^2, \text{sp}^3$
19. Which of the set of species have same hybridization state but different shapes:-
 (1) $\text{NO}_2^+, \text{NO}_2, \text{NO}_2^-$ (2) $\text{ClO}_4^-, \text{SF}_4, \text{XeF}_4$
 (3) $\text{NH}_4^+, \text{H}_3\text{O}^+, \text{OF}_2$ (4) $\text{SO}_4^{2-}, \text{PO}_4^{3-}, \text{ClO}_4^-$
20. The bonds present in N_2O_5 are :
 (1) Only ionic (2) Covalent & coordinate
 (3) Only covalent (4) Covalent & ionic
21. The correct statement for the reaction:

$$\text{NH}_3 + \text{H}^+ \longrightarrow \text{NH}_4^+$$

 (1) Hybridisation state is changed (2) Bond angle increases
 (3) NH_3 act as a Lewis acid (4) Regular geometry becomes irregular
22. The correct order of decreasing polarisability of ions is :
 (1) $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$ (2) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$
 (3) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ (4) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
23. CCl_4 is more covalent than LiCl because :
 (1) There is more polarization of Cl in CCl_4 (2) There is more polarization of Cl in LiCl
 (3) CCl_4 has more weight (4) All of the above
24. An ionic compound $\text{A}^+ \text{B}^-$ is most likely to be formed when -
 (1) Ionization energy of A is low (2) Electron affinity of B is high
 (3) Electron affinity of B is low (4) Both (1) and (2)
25. Which of the following statements regarding HClO_3 is/are correct.
 (1) oxidation state of chlorine is +5 (2) it has two $\text{p}\pi\text{-d}\pi$ bonds
 (3) it has two type of Cl-O bond (4) all of these
26. The correct sequence of increasing covalent character is represented by -
 (1) $\text{BeCl}_2 < \text{NaCl} < \text{LiCl}$ (2) $\text{NaCl} < \text{LiCl} < \text{BeCl}_2$ (3) $\text{BeCl}_2 < \text{LiCl} < \text{NaCl}$ (4) $\text{LiCl} < \text{NaCl} < \text{BeCl}_2$
27. Correct order of bond length is :
 (1) $\text{N}-\text{H} > \text{P}-\text{H} > \text{Sb}-\text{H}$ (2) $\text{N}-\text{H} < \text{P}-\text{H} < \text{Sb}-\text{H}$ (3) $\text{P}-\text{H} > \text{N}-\text{H} > \text{Sb}-\text{H}$ (4) $\text{Sb}-\text{H} > \text{N}-\text{H} > \text{P}-\text{H}$
28. Which of the following species contains three bond pairs and one lone pair around the central atom ?
 (1) NH_2^- (2) PCl_3 (3) H_2O (4) BF_3



29. During change of O_2 to O_2^- ion, the electron adds in which one of the following orbitals?
 (1) $\sigma^* 2p_z$ orbital (2) $\sigma 2p_z$ orbital
 (3) $\pi^* 2p_x / \pi^* 2p_y$ orbital (4) $\pi 2p_x / \pi 2p_y$ orbital
30. The number of S-S bonds in sulphur trioxide trimer (S_3O_9) is :
 (1) 3 (2) 0 (3) 1 (4) 2

Practice Test-1 (IIT-JEE (Main Pattern))

OBJECTIVE RESPONSE SHEET (ORS)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22	23	24	25	26	27	28	29	30
Ans.										

PART - II : NATIONAL STANDARD EXAMINATION IN CHEMISTRY (NSEC) STAGE-I

- Which of the following has maximum bond energy ? [NSEC-2000]
 (A) O_2^- (B) O_2^+ (C) O_2 (D) O_2^{2-}
- The hybrid orbital of the central atom in AlF_4^- is : [NSEC-2000]
 (A) sp (B) sp^2 (C) sp^3 (D) sp^3d
- The paramagnetism of oxygen is best explained by [NSEC-2000]
 (A) MO theory (B) Valence Bond Theory
 (C) VSEPR theory (D) Lewis dot structure
- Which of the following compounds has both ionic and covalent bonding ? [NSEC-2000]
 (A) NaBr (B) $Ba(CN)_2$ (C) PCl_3 (D) CH_3CH_2OH
- Which of the following molecule is non-polar ? [NSEC-2000]
 (A) CO (B) SO_2 (C) SO_3 (D) H_2O
- In ammonia H-N-H bond angle is [NSEC-2001]
 (A) 106.7° (B) 104.5° (C) 109.5° (D) 120°
- The cyanide ion, CN^- and the nitrogen molecule, N_2 are isoelectronic. However, in contrast to CN^- , N_2 is chemically inert due to : [NSEC-2001]
 (A) unsymmetrical electron distribution
 (B) low bond energy
 (C) absence of bond polarity
 (D) presence of a greater number of electrons in bonding orbital
- Valence bond theory was proposed by : [NSEC-2001]
 (A) Heitler and London (B) Slater and Mullikan (C) Pauling (D) Rutherford
- The maximum possible number of hydrogen bonds a water molecule can form is : [NSEC-2001]
 (A) 2 (B) 3 (C) 4 (D) 1
- Hydrogen fluoride is liquid at room temperature because of [NSEC-2002]
 (A) covalent nature (B) hydrogen bonding (C) van der Waals force (D) strong ionic bond.
- Overlapping of the atomic orbitals results in the formation of a covalent bond of the type [NSEC-2002]
 (A) sigma (σ) (B) pi (π) (C) sigma or pi (D) sigma and pi.





12. Which of the following pairs has the strongest hydrogen bonding between themselves ? [NSEC-2002]
 (A) SiH_4 and SiF_4 (B) CH_4 and CH_3OH
 (C) CH_3COCH_3 and CHCl_3 (D) HCO_2H and $\text{CH}_3\text{CO}_2\text{H}$
13. SF_6 molecule has geometry. [NSEC-2002]
 (A) pyramidal (B) bi-pyramidal (C) tetrahedral (D) octahedral
14. Isosteres are compounds having similar geometry and isoelectronic species are species having the same number of electrons. The pair of species CO_2 and NO_2^+ is [NSEC-2003]
 (A) isosteric and isoelectronic (B) isosteric, but not isoelectronic
 (C) isoelectronic, but not isosteric (D) neither isosteric nor isoelectronic.
15. Considering z-axis to be the internuclear axis, the combination of orbitals on Li and Cl atoms respectively, that can lead to a stable sigma bond [NSEC-2003]
 (A) $2s$ and $3p_y$ (B) $1s$ and $3p_y$ (C) $1s$ and $3p_z$ (D) $2s$ and $3p_z$.
16. The number of hydrogen bonds formed by each H_2O molecule in an ice crystal is : [NSEC-2003]
 (A) 6 (B) 4 (C) 2 (D) 3
17. The structures of AlCl_3 and PCl_3 can be described as [NSEC-2004]
 (A) both planar (B) both pyramidal
 (C) planar and pyramidal respectively (D) pyramidal and planar respectively.
18. Bond orders of NO and NO^+ are respectively [NSEC-2004]
 (A) 2.5 and 3 (B) 2 and 4 (C) 3.5 and 2.5 (D) 4 and 2
19. The molecule with non-zero dipole moment is [NSEC-2004]
 (A) BF_3 (B) PCl_3 (C) SiCl_4 (D) XeF_4
20. The molecular orbital with highest energy in a nitrogen molecule is [NSEC-2004]
 (A) σ_{2p} (B) π_{2p} (C) σ^*_{2p} (D) π^*_{2p}
21. Which of the following has zero dipole moment ? [NSEC-2005]
 (A) NH_3 (B) NF_3 (C) BF_3 (D) CHCl_3 .
22. According to VSEPR theory the shape of SF_4 molecule is (atomic number of sulphur=16) [NSEC-2005]
 (A) squashed tetrahedral (B) square pyramidal
 (C) tetrahedral (D) square planar.
23. In which of the following cases, does breaking of the covalent bond take place? [NSEC-2005]
 (A) boiling of water (B) melting of KCN (C) boiling of CF_4 (D) melting of SiO_2 .
24. The hybrid orbitals used by chlorine in ClF_3 molecule are of the type [NSEC-2005]
 (A) sp^3 (B) sp^2 (C) sp^2d (D) sp^3d .
25. In solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, copper is coordinated to [NSEC-2005]
 (A) one water molecule (B) three water molecules
 (C) five water molecules (D) four water molecules.
26. Which of the following contain maximum number of electrons in the antibonding molecular orbital? [NSEC-2005]
 (A) O_2^{2-} (B) O_2 (C) O_2^- (D) O_2^+ .
27. Identify the molecule that has more covalent bond character. [NSEC-2005]
 (A) Na_2S (B) MgCl_2 (C) NaH (D) SnCl_4
28. The non-linear molecule is [NSEC-2006]
 (A) SO_2 (B) CO_2 (C) HCN (D) C_2H_2 .
29. In the ammonium ion, [NSEC-2006]
 (A) the four hydrogens are at the corners of a square
 (B) all bonds are ionic
 (C) the nitrogen atom carries a formal charge
 (D) all bonds are co-ordinate ones.



- 30.* Lewis dot structures of compounds of representative elements normally follow the octet rule. Which of the following does not obey the octet rule ? [NSEC-2006]
 (A) CO_3^{2-} (B) O_3 (C) SO_2 (D) I_3^-
31. Trisilylamine $\ddot{\text{N}}(\text{Si}(\text{CH}_3)_3)_3$ is [NSEC-2006]
 (A) acidic (B) basic (C) neutral (D) amphoteric.
32. Which of the following compounds of phosphorus does not have resonating structures ? [NSEC-2006]
 (A) H_3PO_2 (B) H_3PO_3 (C) H_3PO_4 (D) $(\text{CH}_3)_2\text{HPO}_3$.
33. In thiosulphuric acid $\text{H}_2\text{S}_2\text{O}_3$, the oxidation states of sulphur atoms are [NSEC-2006]
 (A) + II, + II (B) 0, + IV (C) + I, + III (D) -II, + VI.
34. Which type of bond exists between the two boron atoms in a diborane molecule ? [NSEC-2007]
 (A) 2-Center-2electron (B) 3-Center-2-electron
 (C) 3-Center-3-electron (D) 4-Center-4-electron
35. According to Molecular Orbital Theory, the oxygen molecule is- [NSEC-2007]
 (A) Diamagnetic (B) Paramagnetic (C) Ferromagnetic (D) non magnetic
36. Which of the following has zero dipole moment? [NSEC-2007]
 (A) NH_3 (B) NF_3 (C) BF_3 (D) CHCl_3
37. Which of the following pairs is a Lewis acid & a Lewis base ? [NSEC-2007]
 (A) Cl^- & Ag^+ (B) NH_3 & BF_3 (C) SO_4^{2-} & HSO_4^- (D) H^+ & OH^-
38. Assuming a Lewis structure for SO_2 in which all the atoms obey the octet rule, the formal charge on S is: [NSEC-2007]
 (A) +1 (B) 0 (C) +2 (D) 2
39. In which of the following pairs are both molecules polar ? [NSEC-2008]
 (A) O_2 and H_2O (B) BF_3 and PCl_3 (C) SO_2 and SCl_2 (D) CS_2 and NO_2
40. The maximum possible number of hydrogen bonds a water molecule can form is [NSEC-2008]
 (A) 2 (B) 4 (C) 3 (D) 1
41. Which of the following has the highest bond order ? [NSEC-2008]
 (A) O_2 (B) O_2^+ (C) O_2 (D) O_2^{2-}
42. The hybridization of the atomic orbitals of sulphur in SO_3 , SO_4^{2-} and SF_4 are respectively: [NSEC-2009]
 (A) sp , sp^3 , sp^2 (B) sp , sp^2 , sp^3d (C) sp^2 , sp , sp^3 (D) sp^2 , sp^3 , sp^3d
43. The molecule which does not have a net dipole moment is – [NSEC-2009]
 (A) H_2O (B) NH_3 (C) BF_3 (D) BrF_5
44. The molecular geometry for ammonia is – [NSEC-2009]
 (A) saw horse (B) trigonal planar (C) tetrahedral (D) pyramidal
45. The sequence of molecular orbitals for the carbide ion (C_2^{2-}) is – [NSEC-2009]
 (A) $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\pi 2p^4$
 (B) $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\pi 2p^4$ $\sigma 2p^2$
 (C) $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\pi 2p^4$ $\sigma 2p^2$ $\pi^* 2p^2$
 (D) $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$ $\sigma^* 2s^2$ $\pi 2p^4$ $\sigma 2p^2$ $\pi^* 2p^4$
46. The structure of SF_4 is [NSEC-2010]
 (A) Trigonal bipyramidal (B) Square planar
 (C) Squashed Tetrahedral (D) Octahedral
47. The pair which contains both molecules polar is [NSEC-2010]
 (A) O_2 & H_2O (B) CO_2 & PCl_3 (C) SO_2 & SCl_2 (D) CS_2 & NO_2
48. The species that contains maximum number of electrons in the antibonding molecular orbitals is : [NSEC-2010]
 (A) O_2^{2-} (B) O_2 (C) O_2^- (D) O_2^+



49. The compound that has the highest ionic character associated with the X-Cl bond is [NSEC-2010]
(A) PCl_5 (B) BCl_3 (C) CCl_4 (D) SiCl_4
50. According to VSEPR theory the shape of IF_5 molecule will be [NSEC-2011]
(A) tetrahedral (B) trigonal bipyramid (C) square pyramid (D) trigonal planar
51. The formal charges on the atoms underlined are [NSEC-2011]
 $\text{C}_6\text{H}_5 - \underline{\text{C}} \equiv \underline{\text{N}} - \underline{\text{O}}$
(A) C = 0, N = -1, O = +1 (B) C = -1, N = +1, O = -1
(C) C = 0, N = +1, O = -1 (D) C = +1, N = 0, O = -1
52. The compound that **does not** have a π bond is : [NSEC-2011]
(A) SO_2 (B) SF_6 (C) O_2 (D) SO_3
53. The C-O-C bond angle in dimethylether is [NSEC-2011]
(A) $109^\circ 28'$ (B) 110° (C) 120° (D) 180°
54. The relative basic strengths of NH_3 , CH_3NH_2 and NF_3 are in the order : [NSEC-2012]
(A) $\text{CH}_3\text{NH}_2 > \text{NH}_3 > \text{NF}_3$ (B) $\text{NH}_3 > \text{CH}_3\text{NH}_2 > \text{NF}_3$
(C) $\text{NF}_3 > \text{CH}_3\text{NH}_2 > \text{NH}_3$ (D) $\text{CH}_3\text{NH}_2 > \text{NF}_3 > \text{NH}_3$
55. The molecule that has maximum covalent character : [NSEC-2012]
(A) NaH (B) Na_2S (C) CaCl_2 (D) SnCl_4
56. Which of the following compounds has zero dipole moment ? [NSEC-2013]
(A) NH_3 (B) NF_3 (C) BF_3 (D) CHCl_3
57. Which one of the following is not a valid structure for dinitrogen oxide ? [NSEC-2013]
 $\begin{array}{ccc} \ddot{\text{N}} \equiv \ddot{\text{N}} & \ddot{\text{N}} \equiv \ddot{\text{O}} & \ddot{\text{N}} \equiv \text{N} \equiv \ddot{\text{O}} \\ \text{(I)} & & \text{(II)} \\ \ddot{\text{N}} \equiv \text{N} & \text{---} & \ddot{\text{O}}: \\ \text{(III)} & & \text{(IV)} \end{array}$
 (A) I (B) II (C) III (D) IV
58. The bond order of NO^+ ion is : [NSEC-2013]
(A) 1 (B) 2 (C) 2.5 (D) 3
59. Which of the following molecular structures is NOT possible ? [NSEC-2013]
(A) OF_2 (B) SF_2 (C) OF_4 (D) SF_4
60. The molecule having the highest dipole moment is [NSEC-2014]
(A) CO_2 (B) CH_4 (C) NH_3 (D) NF_3
- 61.* The species which has triangular planar geometry is [NSEC-2014]
(A) NF_3 (B) NO_3^- (C) AlCl_3 (D) SbH_3
62. The correct order of stability for the following species is [NSEC-2014]
(A) $\text{Li}_2 < \text{He}_2^+ < \text{O}_2^+ < \text{C}_2$ (B) $\text{C}_2 < \text{O}_2^+ < \text{Li}_2 < \text{He}_2^+$
(C) $\text{He}_2^+ < \text{Li}_2 < \text{C}_2 < \text{O}_2^+$ (D) $\text{O}_2^+ < \text{C}_2 < \text{Li}_2 < \text{He}_2^+$
63. The hybridization of boron in the stable borane having the lowest molecular weight is – [NSEC-2014]
(A) sp^2 (B) sp^3 (C) sp (D) sp^3d
64. For SF_4 , the molecular geometry and hybridization of the central atom respectively are: [NSEC-2014]
(A) Square planar, dsp^2 (B) Tetrahedral, sp^3
(C) Seesaw, sp^3d (D) Square pyramid, sp^3d
65. The species that cannot exist is [NSEC-2014]
(A) SiF_6^{2-} (B) BF_6^{3-} (C) SF_6 (D) AlF_6^{3-}
66. The species having highest bond energy is [NSEC-2015]
(A) O_2 (B) O_2^+ (C) O_2^- (D) O_2^{2-}



67. The structure of a molecule of $N(\text{SiMe}_3)_3$ is [NSEC-2015]
 (A) Pyramidal with angle close to 110° (B) T-shaped with angle 90°
 (C) Bent T-shaped with angle close to 89° (D) Trigonal planar with bond angle close to 120°

68. The order of $p\pi-d\pi$ interaction in the compounds containing bond between Si/P/S/Cl and oxygen is in the order [NSEC-2015]
 (A) $P > \text{Si} > \text{Cl} > \text{S}$ (B) $\text{Si} < P < \text{S} < \text{Cl}$ (C) $\text{S} < \text{Cl} < P < \text{Si}$ (D) $\text{Si} > P > \text{S} > \text{Cl}$

69. Which one of the following information about the compounds is correct ? [NSEC-2016]

Compounds	Oxidation state of P	No. of P-OH bonds	No. of P-H bonds	No. of P=O bonds
[I] H_3PO_2 Hypophosphorous acid	1+	2	1	0
[II] $\text{H}_4\text{P}_2\text{O}_5$ pyrophosphorous acid	3+	2	2	2
[III] $\text{H}_4\text{P}_2\text{O}_6$ Hypophosphoric acid	4+	2	2	2
[IV] $\text{H}_4\text{P}_2\text{O}_7$ pyrophosphoric acid	5+	3	1	4

(A) I

(B) III

(C) IV

(D) II

70. The pair that is isostructural (i.e. having the same shape and hybridization) is [NSEC-2016]
 (A) NF_3 and BF_3 (B) BF_4^- and NH_4^+ (C) BCl_3 and BrCl_3 (D) NH_3 and NO_3^-

71. Number of P-S single bonds and P-S double bonds ($\text{P}=\text{S}$) in P_4S_{10} are respectively [NSEC-2016]
 (A) 10, 6 (B) 16, 0 (C) 14, 2 (D) 12, 4

72. Which of the following compounds contain 3-centered 2-electron bonding ? [NSEC-2016]
 (i) $[\text{BeF}_2]_n$ (ii) $[\text{Be}(\text{CH}_3)_3]_n$ (iii) $[\text{BeCl}_2]_n$ (iv) $[\text{BeH}_2]_n$
 (A) (i) and (ii) (B) (ii) and (iii) (C) (ii) and (iv) (D) (iii) and (iv)

73. In ammonia the bond angle is $107^\circ 48'$ while in SbH_3 the bond angle is about $91^\circ 18'$. The correct explanation among the following is/are [NSEC-2016]
 (A) The orbitals of Sb used for the formation of Sb-H bond are almost pure p-orbitals.
 (B) Sb has larger size compared to N.
 (C) Sb has more metallic character than N.
 (D) All the statements are correct.

74. Assuming that Hund's rule is violated by the diatomic molecule B_2 , its bond order and magnetic nature will be respectively [NSEC-2016]
 (A) 1, diamagnetic (B) 1, paramagnetic (C) 2, diamagnetic (D) 2, paramagnetic

75. In the Lewis structure of ozone (O_3), the formal charge on the central oxygen atom is [NSEC-2017]
 (A) +1 (B) -1 (C) 0 (D) -2

76. Which of the following represents the correct order of dipole moment? [NSEC-2017]
 (A) $\text{NH}_3 > \text{NF}_3 > \text{H}_2\text{O}$ (B) $\text{NH}_3 > \text{H}_2\text{O} > \text{NF}_3$ (C) $\text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3$ (D) $\text{H}_2\text{O} > \text{NF}_3 > \text{NH}_3$

77. Which of the following has the shortest bond length ? [NSEC-2017]
 (A) O_2 (B) O_2^- (C) O_2^+ (D) O_2^{2-}

- 78.* Which of the following is not paramagnetic? [NSEC-2018]
 (A) S^{2-} (B) N^{2-} (C) O^{2-} (D) NO

79. In which of the following, all the bond lengths are not the same ? [NSEC-2018]
 I. IF_4^+ II. BF_4^- III. SF_4 IV. TeCl_4
 (A) I, II, IV (B) II, III, IV (C) I, III, IV (D) I, II, III

80. Among the following pairs, the one in which both the compounds as pure liquids can show significant auto ionization is [NSEC-2018]
 (A) H_2O and H_2S (B) BrF_3 and ICl_3 (C) PF_5 and PCl_5 (D) HF and HCl



81. Which among the following is nonlinear ?

[NSEC-2018]

- (A) N_3^- (B) ClF_2^- (C) Br_3^- (D) BrCl_2^+

82. The most stable Lewis structure of N_2O is

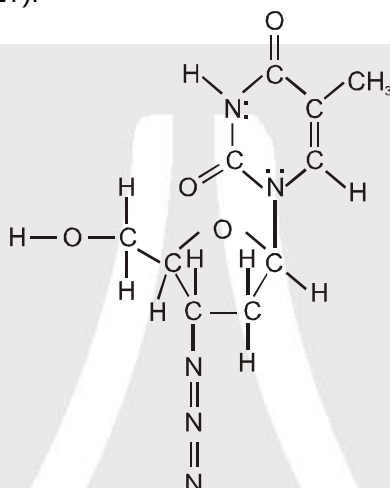
[NSEC-2018]

- (A) $\text{:}\ddot{\text{O}}=\ddot{\text{N}}=\ddot{\text{N}}\text{:}$ (B) $\text{:}\text{N}=\text{O}=\ddot{\text{N}}\text{:}$ (C) $\text{:}\ddot{\text{N}}-\text{N}=\text{O}\text{:}$ (D) $\text{:}\ddot{\text{O}}-\text{N}\equiv\text{N}\text{:}$

PART - III : HIGH LEVEL PROBLEMS (HLP)

SUBJECTIVE QUESTIONS

- Draw the structure of the following compound and identify the hybridisation of the central atom, also count the Sigma and π -bond.
(i) XeO_2F_2 (ii) PF_3Cl_2 (iii) NH_2OH (iv) Anion of $\text{PCl}_5(\text{s})$ (v) P_4
- One of the first drugs to be prepared for use in treatment of acquired immuno deficiency syndrome (AIDS) was azidothymidine (AZT).



- How many carbon atoms have sp^3 hybridisation ?
 - How many carbon atoms have sp^2 hybridisation ?
 - How many nitrogen atoms (central atom not terminal) have sp hybridisation ?
 - How many π bonds are in the molecule ?
- Draw structures for the polymeric $(\text{BeH}_2)_n$ and $(\text{BeCl}_2)_n$. Explain in brief why the hydride bridge in $(\text{BeH}_2)_n$ is considered to be electron deficient but not the halide bridge in $(\text{BeCl}_2)_n$?
 - Explain the structure, hybridisation and oxidation state of S in sulphuric acid, Marshall's acid, Caro's acid and oleum.
 - Give the number of characteristic bond(s) found in the various oxy-acids of phosphorus as given below.
(P) Number of P-O-P bond(s) in cyclotrimetaphosphoric acid.
(Q) Number of P-P bond(s) in hypophosphoric acid.
(R) Number of P-H bond(s) in hypophosphorus acid.
(S) Number of P-OH bond(s) in pyrophosphoric acid.

P	Q	R	S



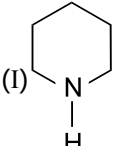
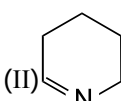
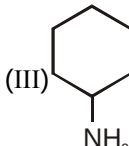
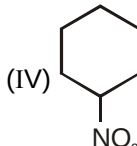
ONLY ONE OPTION CORRECT TYPE

9. Example of super octet molecule is :
 (A) SF_6 (B) PCl_5 (C) IF_7 (D) All of these
10. Correct order of bond energy of C–O bond is :
 (A) $\text{CO}_3^{2-} > \text{CO}_2 > \text{CO}$ (B) $\text{CO}_2 > \text{CO} > \text{CO}_3^{2-}$
 (C) $\text{CO} > \text{CO}_2 > \text{CO}_3^{2-}$ (D) None of these.
11. Arrange the following ions in the order of decreasing X–O bond length where X is the central atom
 (A) ClO_4^- , SO_4^{2-} , PO_4^{3-} , SiO_4^{4-} (B) SiO_4^{4-} , PO_4^{3-} , SO_4^{2-} , ClO_4^-
 (C) SiO_4^{4-} , PO_4^{3-} , ClO_4^- , ClO_4^{2-} (D) SiO_4^{4-} , SO_4^{2-} , PO_4^{3-} , ClO_4^-
12. Which one (s) of the following structures cannot represent resonance forms for N_2O (diamagnetic)?
 (a) $\text{:}\ddot{\text{N}} = \text{N} = \ddot{\text{O}}\text{:}$ (b) $\text{:}\ddot{\text{N}} \equiv \text{N} - \ddot{\text{O}}\text{:}$ (c) $\text{:}\ddot{\text{N}} - \text{N} \equiv \text{O}\text{:}$
 (d) $\text{:}\ddot{\text{N}} = \text{O} = \ddot{\text{N}}\text{:}$ (e) $\text{:}\ddot{\text{N}} = \text{N} = \ddot{\text{O}}\text{:}$
 (A) a and c (B) c, d and e (C) d and e (D) c and d
13. Which of the following overlaps gives a σ bond with x as internuclear axis?
 (A) p_z and p_z (B) s and p_z (C) s and p_x (D) $d_{x^2-y^2}$ and $d_{x^2-y^2}$
14. The structure of F_2SeO is analogous to :
 (A) SO_3 (B) ClO_3^- (C) XeO_3 (D) (B) and (C) both
15. In which of the following N is in the sp^2 hybridisation state
 (A) $(\text{CH}_3)_3\text{N}$ (B) $\text{CH}_2 = \text{NH}$ (C) CH_3CN (D) NO_2^+
16. In H_2SO_4 molecule
 (A) S-atom is sp^3d^2 hybridised and there are 4 lone pair electrons on 'O' atoms.
 (B) S-atom is sp^3 hybridised and there are no lone pair of electrons in the molecule.
 (C) S-atom is sp^3 hybridised and there are 8 lone pair of electrons on the 'O' atoms.
 (D) S-atom is sp^2 hybridised and these are 8 lone pair of electrons 'O' atoms
17. In which of the following pairs hybridisation of the central atoms are different ?
 (A) ClF_3 , ClF_3O (B) ClF_3O , ClF_3O_2 (C) $[\text{ClF}_2\text{O}]^+$, $[\text{ClF}_4\text{O}]^-$ (D) $[\text{ClF}_4\text{O}]^-$, $[\text{XeOF}_4]$
18. All the following species have all their bond lengths identical except :
 (A) AsF_3 (B) AsF_4^- (C) AsF_4^+ (D) AsF_6^-
19. Which one has highest bond angle.
 (A) NH_3 (B) PH_3 (C) H_2O (D) CH_4
20. In the thiocyanate ion, SCN^- three resonating structure are possible with the electron-dot method as shown in figure :

$$\begin{array}{ccc} \text{:}\ddot{\text{S}} = \text{C} = \text{N:}^- & \text{:}\ddot{\text{S}}^- - \text{C} \equiv \text{N:} & \text{S}^+ \equiv \text{C} - \text{N}^{2-} \\ \text{(x)} & \text{(y)} & \text{(z)} \end{array}$$

 The decreasing order of % contribution in resonance hybrid is :
 (A) $y > x > z$ (B) $y > z > x$ (C) $z > x > y$ (D) cannot predicted.
21. Which of the following overlappings will result into strongest bond formation. (Consider X to be internuclear axis)
 (A) $1s-1s$ overlap (B) $2p_y-2p_y$ overlap
 (C) $2p_x-2p_z$ overlap (D) $2p_x-2p_x$ overlap
22. Phosphorous penta-chloride in gaseous phase exists as a monomer. In solid state, it exists as PCl_4^+ and PCl_6^- ions. The hybrid state of P-atom in PCl_5 is sp^3d . The hybrid states of P-atoms in PCl_4^+ and PCl_6^- will be :
 (A) sp^3d , sp^3d^2 (B) sp^3 , sp^3d^2 (C) sp^3d^2 , sp^3d (D) sp^3 , sp^3d



23. In which of the following cyclic compound the nitrogen atom is sp^3 hybridised
- (I)  (II)  (III)  (IV) 
- (A) I & III (B) I, II, III (C) III & IV (D) I, III & IV
24. Which of the statements is correct about SO_2 ?
- (A) two σ , two π and no lone pair of electrons (B) two σ and one π
 (C) two σ , two π and one lone pair (D) none of these
25. Arrange the following in the increasing order of deviation from normal tetrahedral angle :
- (A) $P_4 < PH_3 < H_2O$ (B) $PH_3 < H_2O < P_4$ (C) $P_4 < H_2O < PH_3$ (D) $H_2O < PH_3 < P_4$
26. In XeF_2 molecule the angle between two lone pair orbitals is α , the angle between lone pair orbital and bond pair orbital is β and the angle between bond pair orbitals is γ :
- (A) $\alpha = \beta = \gamma$ (B) $\alpha > \beta > \gamma$ (C) $\gamma > \beta > \alpha$ (D) $\gamma > \alpha > \beta$
27. In O_2F_2 , which of the following statement is incorrect.
- (A) O–F bond length in O_2F_2 is longer than O–F bond length in OF_2 .
 (B) The O.N. of oxygen in O_2F_2 is +1.
 (C) The O–O bond length in O_2F_2 is shorter than O–O bond length in H_2O_2 .
 (D) None of these
28. Identify the species containing Banana bonds
- (A) $(BeH_2)_n$ (B) BF_3 (C) $(AlCl_3)_2$ (D) $(BeCl_2)_n$
29. In the coordinate valency
- (A) Electrons are equally shared by the atoms
 (B) Electrons of one atom are shared between two atoms
 (C) Hydrogen bond is formed
 (D) Electrons are completely donated to other atom with no sharing
30. What is the nature of the bond between B and O in $(C_2H_5)_2OBH_3$.
- (A) Covalent (B) Co-ordinate covalent (C) Ionic bond (D) Banana shaped bond
31. Which of the following species does not contain N–N covalent bond ?
- (A) N_2O_3 (B) $N_2O_2^{2-}$ (C) N_2O_5 (D) N_2O_4
32. Compounds in which oxidation state of at least one oxygen is (-1) :
- (1) H_2O_2 (2) O_2F_2 (3) H_2SO_5 (4) CrO_5
 (5) $H_2S_2O_8$ (6) $[Na_2B_2H_4O_8]$
 (A) 1,2,3,5 (B) 1,3,4,5,6 (C) 1,3,4,5 (D) 1,2,3,4,5,6
33. Which of the following statements is incorrect ?
- (A) Among O_2^+ , O_2 and O_2^- the stability decreases as $O_2^+ > O_2 > O_2^-$
 (B) He_2 molecule does not exist as the effect of bonding and anti-bonding orbitals cancel each other.
 (C) C_2 , O_2^{2-} and Li_2 are diamagnetic
 (D) In F_2 molecule, the energy of σ_{2p_z} is more than π_{2p_x} and π_{2p_y}
34. Which of the following statement is incorrect ?
- (A) During N_2^+ formation, one electron is removed from the bonding molecular orbital of N_2 .
 (B) During O_2^+ formation, one electron is removed from the antibonding molecular orbital of O_2 .
 (C) During O_2^- formation, one electron is added to the bonding molecular orbital of O_2 .
 (D) During CN^- formation, one electron is added to the bonding molecular orbital of CN .
35. Which the following molecules / species have identical bond order and same magnetic properties ?
- (I) O_2^+ ; (II) NO ; (III) N_2^+
 (A) (I) , (II) only (B) (I) and III only (C) (I), (II) and (III) (D) (II) and (III) only



36. Which of the following has the minimum heat of dissociation of $N \rightarrow B$ bond ?
 (A) $[(CH_3)_3N \rightarrow BF_3]$ (B) $[(CH_3)_3N \rightarrow B(CH_3)F_2]$
 (C) $[(CH_3)_3N \rightarrow B(CH_3)_2F]$ (D) $[(CH_3)_3N \rightarrow B(CH_3)_3]$
37. Which of the following does not have coordinate bond ? (Consider that every atom should follow octet rule and H-atom should follow duet rule).
 (A) SO_2 (B) HNO_2 (C) H_2SO_3 (D) HNO_3
38. Gaseous SO_3 molecule
 (A) is planar triangular in shape with three σ -bonds from sp^2 -p overlap and three π -bonds formed by one $p\pi$ - $p\pi$ overlap and two $p\pi$ - $d\pi$ overlap.
 (B) is a pyramidal molecule with one double bond and two single bonds
 (C) planar triangular in shape with two double bonds between S and O and one single bond
 (D) is planar triangular in shape with three σ bonds from sp^2 -p overlap and three π -bonds formed by two $p\pi$ - $p\pi$ overlap and one $p\pi$ - $d\pi$ overlap.
39. Consider the following statements.
 S_1 : Fluorine does not form any polyhalide because it does not have d-orbitals in valence shell.
 S_2 : In ClF_3 , the three lone pairs of electrons occupy the equatorial position.
 S_3 : In B_2 and N_2 molecules mixing of s- and p- atomic orbitals takes place.
 Of these statements :
 (A) S_1 , S_2 and S_3 are correct (B) S_1 and S_2 are correct
 (C) S_1 and S_3 are correct (D) S_2 and S_3 are correct
40. When N_2 goes to N_2^+ , the N-N bond distance ... and when O_2 goes to O_2^+ , the O-O bond distance....
 (A) Decrease, Increases (B) Increases, Decrease
 (C) Increases, Increases (D) None of these
41. According to Molecular orbital theory which of the following is correct ?
 (A) LUMO level for C_2 molecule is σ_{2p_x} orbital (B) In C_2 molecules both the bonds are π bonds
 (C) In C_2^{2-} ion there is one σ and two π bonds (D) All the above are correct
42. Which of the following is incorrect ?
 (A) The O-O bond length in H_2O_2 is larger than that in O_2F_2 .
 (B) The O-O bond length in H_2O_2 is very slightly smaller than in O_2^{2-} ion.
 (C) O_2^- and O_2^+ species are paramagnetic and have same number of unpaired electrons.
 (D) None
43. A simplified application of MO theory to the hypothetical 'molecule' OF would give its bond order as :
 (A) 2 (B) 1.5 (C) 1.0 (D) 0.5
44. Which one of the following oxides is expected to exhibit paramagnetic behaviour
 (A) CO_2 (B) SO_2 (C) ClO_2 (D) SiO_2
45. Which of the following is correct :
 (A) N-O bond length in NO gaseous molecule will be greater than in NOCl gaseous molecule.
 (B) Carbon-carbon bond length in CaC_2 will be more than in C_2H_4
 (C) O-O bond length in KO_2 will be more than in Na_2O_2 .
 (D) All the four hydrogen atoms in CH_4 are not coplanar
46. Consider the following statements ;
 (I) The hybridisation found in cation of solid PCl_5 is sp^3 .
 (II) In AB_2L_2 type the BAB bond angle is always greater than the normal tetrahedral bond angle.
 (III) In ClO_3^- , NH_3 and XeO_3 , the hybridisation and the number of lone pairs on the central atoms are same.
 (IV) In P_4 molecule, there are six P-P bonds and four lone pairs of electrons.
 of these statements :
 (A) I, II and III are correct only (B) I, III and IV are correct only
 (C) III and IV are correct only (D) All are correct



47. (a) There are only 12 bonding electrons available in one molecule of diborane.
 (b) $B_3N_3H_6$ is an electron deficient compound.
 (c) Al_2Cl_6 sublimes on heating and give $AlCl_3$ vapours at high temperature.
 (d) In $Si_2O_7^{6-}$ anion, one oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron.
 (A) T F T T (B) F T F F (C) T F T F (D) F T F T
48. The point of dissimilarity between $(SO_3)_3$ and $(HPO_3)_3$ (cyclic trimers) is.
 (A) having six membered ring. (B) having central atom in same hybridization
 (C) having planar ring. (D) being isoelectronic.
49. Which of the following statements is/are true about hypophosphoric acid.
 (A) Oxidation state of phosphorus atoms present in hypophosphoric acid is +3
 (B) Phosphorous atom is present in 2 different oxidation state which are +3, +5
 (C) Oxidation state of phosphorous in hypophosphoric acid is +4
 (D) Oxidation state of phosphorus in hypophosphoric acid is +5.
50. The gaseous HX molecule has a measured dipole moment of 4.0 D, which indicates that it is a very polar molecule. The separation between the nuclei in this molecule is 2.67×10^{-8} cm then the percentage ionic character in HX molecule is :
 (A) 78% (B) 31.25% (C) 50.25% (D) None of these
51. Which of the following models best describes the bonding within a layer of the graphite structure ?
 (A) metallic bonding (B) ionic bonding
 (C) non-metallic covalent bonding (D) van der Waals forces
52. Which of the following compounds would have significant intermolecular hydrogen bonding ?
 (A) HCl (B) H_2S (C) chloral hydrate (D) CH_3OH
53. Which of the following statement is not true ?
 (A) CCl_4 has higher boiling point than $CHCl_3$.
 (B) The HF_2^- ion exists in the solid state and in liquid HF solution, but not in dilute aqueous solutions.
 (C) Hydrogen bonding maintains the planar H_3BO_3 units in layers in solid state.
 (D) None of these.
54. S_1 : In the solid $B(OH)_3$ units are hydrogen bonded together into two-dimensional sheets with almost hexagonal symmetry.
 S_2 : Na_2CO_3 is isomorphous with Na_2SO_3 as both have similar formula type.
 S_3 : XeO_3F_2 has one lone pair of electron on central xenon atom.
 S_4 : D_2O has higher boiling point than H_2O
 (A) T F T F (B) T F F T (C) T T F F (D) T T T T
55. Which is correct about D_2O
 (A) Its boiling point is higher than that of $H_2O(l)$ (B) O—D----O bond is stronger than O—H----O bond.
 (C) $D_2O(s)$ sinks in $H_2O(l)$. (D) all the above are correct.
56. Among the following compounds, the correct order of the polarity of the bonds is :
 SbH_3, AsH_3, PH_3, NH_3 .
 (A) $SbH_3 < AsH_3 < PH_3 < NH_3$ (B) $AsH_3 < SbH_3 = PH_3 < NH_3$
 (C) $PH_3 < AsH_3 < SbH_3 < NH_3$ (D) $AsH_3 < PH_3 < SbH_3 < NH_3$
57. Given the species N_2, CO, CN^- and NO^+ . Which of the following statement is incorrect .
 (A) All the species are diamagnetic
 (B) All the species are isoelectronic
 (C) All the species have dipole moment
 (D) All the species are linear
58. CH_3Cl has more dipole moment than CH_3F because :
 (A) electron affinity of chlorine is greater than that of fluorine.
 (B) the charge separation is larger in CH_3Cl compared to CH_3F .
 (C) the repulsion between the bond pairs and non-bonded pairs of electrons is greater in CH_3Cl than CH_3F .
 (D) chlorine has higher electronegativity than fluorine.



59. Select the correct statement.
 (A) The order of Xe–F bond length in various fluorides of Xenon is $\text{XeF}_2 < \text{XeF}_4 < \text{XeF}_6$
 (B) PH_5 can undergo sp^3d hybridisation to have octahedral geometry.
 (C) Dipole moment of CH_3F is greater than that of CH_3Cl
 (D) Increasing strength of hydrogen bonding is $\text{Cl}-\text{H}\cdots\text{Cl} < \text{N}-\text{H}\cdots\text{N} < \text{O}-\text{H}\cdots\text{O} < \text{F}-\text{H}\cdots\text{F}$
60. The pairs of bases in DNA are held together by
 (A) Hydrogen bonds (B) Ionic bonds
 (C) Phosphate groups (D) Deoxyribose groups
61. The bond that determines the secondary structure of a protein is
 (A) Coordinate bond (B) Covalent bond (C) Hydrogen bond (D) Ionic bond
62. Which of the following has highest melting point according to Fajan's rule :
 (A) NaCl (B) MgCl_2 (C) AlCl_3 (D) LiCl
63. An element X occurs in short period having configuration ns^2np^1 . The formula and nature of its oxide is :
 (A) XO_3 , basic (B) XO_3 , acidic (C) X_2O_3 , amphoteric (D) X_2O_3 , basic
64. Which of the following compounds of elements in group IV is expected to be most ionic ?
 (A) PbCl_2 (B) PbCl_4 (C) CCl_4 (D) SiCl_4
65. Which of the following cannot be explained on the basis of Fajan's Rules ?
 (A) Ag_2S is much less soluble than Ag_2O
 (B) $\text{Fe}(\text{OH})_3$ is much less soluble than $\text{Fe}(\text{OH})_2$
 (C) BaCO_3 is much less soluble than MgCO_3
 (D) Melting point of AlCl_3 is much less than that of NaCl
66. **S₁** : AgI is less soluble in water than AgF due to more polarisation of I^- in comparison to F^- ion.
S₂ : Melting point of BaCl_2 is higher than the melting point of BeCl_2 due to greater ionic nature of BaCl_2 .
S₃ : Order of hydrated radii is : $\text{Al}^{3+}(\text{aq}) > \text{Mg}^{2+}(\text{aq}) > \text{Na}^+(\text{aq})$
 (A) T T T (B) T T F (C) T F T (D) F T T
67. Correct order of strength of metallic bond in Li, K, Fe, W
 (A) $\text{W} > \text{Fe} > \text{Li} > \text{K}$ (B) $\text{Fe} > \text{W} > \text{K} > \text{Li}$ (C) $\text{Li} > \text{K} > \text{W} > \text{Fe}$ (D) $\text{K} > \text{Li} > \text{Fe} > \text{W}$
68. The ground state electronic configurations of some elements, P, Q, R, S, and T (these symbols represent the some of the known elements given in the periodic table) are as follows.
 P : $1s^2 2s^2 2p^6 3s^2 3p^2$
 Q : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
 R : $1s^2 2s^2 2p^6 3s^2 3p^1$
 S : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
 T : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$.
 Match the electronic configurations of the elements with the properties given below and select the correct sequence by choosing the correct codes given.
 (i) Element forms a cation which is isoelectronic with P^{3-} .
 (ii) Element which in its compounds can show a maximum oxidation state of +6 and that is coloured too.
 (iii) Element has largest atomic radius and highest first ionisation energy in the respective period.
 (iv) Element which has intermediate value of electronegativity and its oxide forms salts with strong acids and bases.
 (A) Q R T P (B) Q S T R (C) Q R S T (D) P Q R S
69. Increasing order of extent of hydrolysis CCl_4 , MgCl_2 , AlCl_3 , PCl_5 , SiCl_4
 (A) $\text{MgCl}_2 < \text{AlCl}_3 < \text{CCl}_4 < \text{AlCl}_4 < \text{PCl}_5$
 (B) $\text{MgCl}_2 > \text{AlCl}_3 > \text{CCl}_4 > \text{AlCl}_4 > \text{PCl}_5$
 (C) $\text{CCl}_4 < \text{MgCl}_2 < \text{AlCl}_3 < \text{SiCl}_4 < \text{PCl}_5$
 (D) $\text{SiCl}_4 < \text{PCl}_5 < \text{CCl}_4 < \text{MgCl}_2 < \text{AlCl}_3$



MATCH THE COLUMN

70. Match the species given in column-I with the type of hybridisation given in column-II.

	Column-I		Column-II
(A)	I_3^-	(p)	sp^3d
(B)	XeO_3	(q)	sp^3
(C)	$ClOF_3$	(r)	sp^2
(D)	XeF_5^+	(s)	sp^3d^2

71. Match the column :

	Column - I		Column - II
(A)	SO_3 (gas)	(p)	Polar with $p\pi-d\pi$ bonds and identical S–O bond, lengths.
(B)	OSF_4	(q)	One lone pair and $p\pi-d\pi$ bond.
(C)	SO_3F^-	(r)	Non-polar with $p\pi-p\pi$ and $p\pi-d\pi$ bonds. Identical S–O bond lengths.
(D)	$ClOF_3$	(s)	Polar with $p\pi-d\pi$ bond.

72. Match the column :

	Column-I		Column-II
(A)	IF_2^-	(p)	sp^3d
(B)	ClF_3	(q)	polar
(C)	XeO_3F_2	(r)	one of the bond angles is 180°
(D)	SF_4	(s)	one lone pair

SINGLE AND DOUBLE VALUE INTEGER TYPE

73. Number of hypervalent among following are :

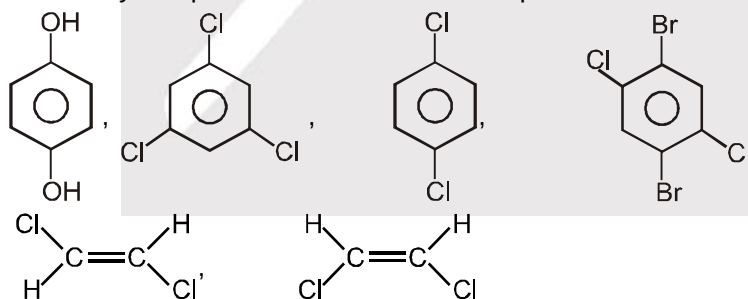
(i) H_2CO_3 , (ii) HNO_3 , (iii) HNO_2 , (iv) H_3PO_4 , (v) H_3PO_3 , (vi) H_3PO_2 , (vii) $HClO_4$, (viii) $HClO_3$, (ix) $HClO_2$, (x) $HClO$, (xi) H_2SO_3 , (xii) H_2SO_4 , (xiii) $H_2N_2O_2$, (xiv) H_2SO_5 , (xv) $H_2S_2O_8$, (xvi) SO_3 , (xvii) SO_2 , (xviii) N_2O_5 , (xix) P_4O_{10}

74. **Compound** **No. of peroxide linkage(s)**

1.	$HXeO_4^-$	—	x
2.	K_3CrO_8	—	y
3.	H_2TiO_4	—	z
4.	$Na_2B_4O_7$	—	w

The value of $x + y + z + w$ is/are

75. How many compound which show $\approx$ zero dipole moment.

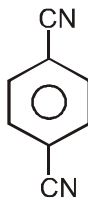


76. A bond X–Y has a dipole moment of 1.8×10^{-29} Cm and a bond length 150 pm. What will be the percentage of ionic character is given bond.



77. Number of non-polar molecule among the following is x and number of planar molecule is y. Give $x \times y$.

BF_3 , CO_2 , SO_2 , PCl_5 , ClF_3 , NH_3 , CH_4 ,



78. Among the following, give the number of correct order of boiling point or melting point.
- | | | | |
|---|---------------|---|---------------|
| (1) $\text{Zn} > \text{Cd} > \text{Hg}$ | Boiling point | (2) $\text{Zn} > \text{Cd} > \text{Hg}$ | Melting point |
| (3) $\text{K} > \text{Ca} > \text{Sc}$ | Boiling point | (4) $\text{Na} > \text{Mg} > \text{Al}$ | Boiling point |
| (5) $\text{Sc} > \text{Mn} > \text{Zn}$ | Melting point | (6) $\text{Pt} > \text{Pd} > \text{Ni}$ | Melting point |
| (7) $\text{Cr} > \text{Mn} < \text{Fe}$ | Melting point | (8) $\text{Ba} > \text{Li} > \text{Na}$ | Melting point |
79. Calculate total number of coordinate bonds in the following molecules.
 (a) PCl_6^- (b) $\text{NH}_3 \cdot \text{BF}_3$ (c) HNO_3 (d) CO
80. Determine number of bonding electrons in H_3SiNCO .

ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

81. Hypervalent compound is(are) :
 (A) SO_3^{2-} (B) PO_4^{3-} (C) SO_4^{2-} (D) ClO_4^-
82. Which of the following overlaps is/are **incorrect** [assuming X-axis to be the internuclear axis] :
 (a) $2p_y + 2p_y \rightarrow \pi$ (b) $2p_z + 2p_z \rightarrow \sigma$ (c) $2p_x + 2p_x \rightarrow \pi$
 (d) $1s + 2p_y \rightarrow \pi$ (e) $2p_y + 2p_z \rightarrow \pi$ (f) $1s + 2s \rightarrow \sigma$
 (A) 'a' & 'b' (B) 'b' & 'd' (C) 'd' & 'f' (D) 'c' & 'e'
83. In which of following, vacant orbital take part in hybridisation :
 (A) BF_3 (B) PCl_6^- (C) BF_4^- (D) $\ddot{\text{N}}\text{H}_3$
84. Which is not true about VSEPR theory
 (A) Lone pair-lone pair repulsion is maximum.
 (B) Lone pair and double bond occupy axial position in trigonal bipyramidal structure.
 (C) More electronegative atoms occupies axial position in trigonal bipyramidal structure.
 (D) Bigger atoms occupy axial positions in trigonal bipyramidal structure.
85. Select the **correct** statement.
 (A) Perxenate ion is $[\text{XeO}_6]^{4-}$ with octahedral geometry.
 (B) XeF_2 is linear molecule with 3 lone pairs (l.p)
 (C) XeOF_4 , XeF_4 , XeO_2F_2 all contains one lone pair only
 (D) None of these
86. Identify the **correct** option(s)
 (A) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$ order of bond angle
 (B) $(\text{CH}_3)_3\text{B}$ is a trigonal planar molecule (not considering the H-atoms on 'C')
 (C) In NH_4Cl 'N' atom is in sp^3d hybridisation
 (D) In S_8 molecule a total of 16 electrons are left on all the 'S' atoms after bonding.
87. Which of the following statements is/are correct for ClO_4^- oxoanion ?
 (A) It does not have any tendency of polymerisation.
 (B) It has strong $p\pi-p\pi$ bonding between chlorine and oxygen.
 (C) All Cl-O bonds are identical and chlorine atom is sp^3 hybridised.
 (D) The chemical bonding takes place in ground state and charge dispersion is more than ClO_3^- oxo anion.
88. Among the following back bonding is exhibited by
 (A) BF_3 (B) $(\text{CH}_3)_3\text{N}$ (C) CH_3NCO (D) SiH_3NCO
89. Which of the following reactions is/are likely to be impossible.
 (A) $(\text{CH}_3)_2\text{O} + \text{BF}_3 \rightarrow$ (B) $(\text{SiH}_3)_2\text{O} + \text{BF}_3 \rightarrow$ (C) $(\text{SiH}_3)_3\text{N} + \text{BF}_3 \rightarrow$ (D) All the above





90. Which of the following statements is/are correct ?
 (A) Out of trimethylamine and trimethylphosphine, trimethylamine has higher dipole moment.
 (B) Out of $(\text{SiH}_3)_2\text{O}$ and $(\text{CH}_3)_2\text{O}$, $(\text{SiH}_3)_2\text{O}$ is more basic.
 (C) C–C bond length (in pm) in C_2 molecule is greater than O–O bond length in O_2 molecule.
 (D) $\text{N}(\text{SiMe}_3)_3$ and BF_3 molecules are isostructural about central atom.
91. In N_2O_3 , the oxidation state of nitrogen atom is
 (A) same for both the nitrogen (B) different for both the nitrogen and they differ by 2.
 (C) one's is +2 and other is +4 (D) No, nitrogen atom is having +3 oxidation state.
92. Which of the following statements is incorrect about P_4O_{10} molecule ?
 (A) Each 'P' atom can be considered to be sp^3 hybridised
 (B) There are four POP bonds in the molecule
 (C) There are two types of P–O bond lengths
 (D) POP angle is 180° .
93. In the following, identify the incorrect statements.
 (A) N_2F_3^+ is a planar at each nitrogen atom
 (B) In F_2 molecule, the energy of $\sigma 2p_z$ is more than $\pi 2p_x$ and $\pi 2p_y$.
 (C) The O–O bond length in H_2O_2 is smaller than in O_2F_2 .
 (D) B_2 , O_2 and F_2 are paramagnetic molecules.
94. In which of the following compound(s), is/are central atom have two different individual oxidation state(s) ?
 (A) C_3O_2 (B) Br_3O_8 (C) $\text{S}_4\text{O}_6^{2-}$ (D) $\text{P}_2\text{O}_7^{4-}$
95. Which of the following statements are correct
 (A) o-hydroxy benzaldehyde is a liquid at room temperature due to intra molecular H-bonding, where as p-hydroxybenzaldehyde is a solid at room temperature.
 (B) Order of boiling point $\text{H}_2\text{O} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
 (C) Order of boiling point $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
 (D) Order of boiling point $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$
96. Which of the following molecules have intermolecular hydrogen bonds ?
 (A) KH_2PO_4 (B) H_3BO_3 (C) $\text{C}_6\text{H}_5\text{CO}_2\text{H}$ (D) CH_3OH
97. Which is **correct** statement :
 (A) Borazine has higher intermolecular force of attraction as compared to benzene.
 (B) The O–O bond length in $\text{O}_2[\text{AsF}_4]$ is shorter than KO_2 .
 (C) The bond angle order in halogen –S– halogen is $\text{OSF}_2 < \text{OSCl}_2 < \text{OSBr}_2$
 (D) None of these
98. The critical temperature of water is higher than that of O_2 because the H_2O molecule has :
 (A) fewer electrons than O_2 (B) two covalent bonds
 (C) V-shape (D) H-bonding
99. Which of the following is/are correct statement(s).
 (A) Increasing covalent character : $\text{NaCl} < \text{MgCl}_2 < \text{AlCl}_3$
 (B) Increasing covalent character : $\text{LiF} < \text{LiCl} < \text{LiBr} < \text{LiI}$.
 (C) Increasing polarizability : $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
 (D) Decreasing ionic nature : $\text{MCl}_3 > \text{MCl}_2 > \text{MCl}$
100. Select the correct statement(s) :
 (A) Among d-block elements, group 12 elements have the least enthalpy of atomisation.
 (B) Cohesive forces in metals are non-directional bonds.
 (C) Group 1 elements have greater strength of metallic bonding than corresponding group 2 elements due to increasing electronegativity on moving left to right in periodic table.
 (D) Hardness of copper being more as compared to potassium can be explained by metallic bond strength.



101. Select the correct statement(s) :
- (A) In s-block elements, melting point generally decreases on moving down the group.
 (B) Band model proposed to explain metallic bonding is based on molecular orbital theory.
 (C) d-block elements get a dip at/near middle of series in enthalpy of atomisation values.
 (D) Electricity conduction in a single layer of graphite can be considered similar to electricity conduction in metals according to electron sea model.
102. Which of the following statements is / are true for the metallic bond ?
- (A) It is an electrical attraction between delocalised electrons and the positive part of the atom.
 (B) Transition metals may use inner d- electrons along with the outer s-electrons for metallic bonding.
 (C) Strength of metallic bond does not depend on the type of hybrid orbitals participating in metallic bonding.
 (D) Strength of metallic bond is inversely proportional to the radius of metallic atom in s-block.

COMPREHENSION

Read the following passage carefully and answer the questions.

Comprehension # 1

The electronic configuration of three elements, A, B and C are given below. Answer the questions 01 to 04 on the basis of these configurations.

A	$1s^2$	$2s^2$	$2p^6$		
B	$1s^2$	$2s^2$	$2p^6$	$3s^2$	$3p^3$
C	$1s^2$	$2s^2$	$2p^6$	$3s^2$	$3p^5$

103. Stable form of A may be represented by the formula :
- (A) A (B) A_2 (C) A_3 (D) A_4
104. Stable form of B may be represented by the formula :
- (A) B (B) B_2 (C) B_3 (D) B_4
105. The molecular formula of the compound formed from B and C will be
- (A) BC_3 (B) B_2C (C) BC_5 (D) A and C both
106. The bond between B and C will be
- (A) Ionic (B) Covalent (C) Hydrogen (D) Coordinate

Comprehension # 2

$Na(BH_4)$ is ionic compound containing BH_4^- tetrahydrido borate ion and in solid state $NaBH_4$ has sodium chloride structure. Not all tetrahydrido borates are ionic the beryllium, aluminium and transition metal borohydrides become increasingly covalent and volatile. In these type of tetrahydrido borate the BH_4^- ion form covalent bond with metal ion. One or more H atoms in BH_4^- act as a bridging and bond to metal forming a 3 center bond with 2 electron shared by 3 atoms. The BH_4^- is usually in that it may form 1, 2 or 3 such 3 center bonds to the metal ion when form covalent bond.

107. $Li[AlH_4]$ is used as a reducing agent in many reactions and it is prepared by excess LiH and $AlCl_3$. select incorrect statements about $Li[AlH_4]$
- (A) hybridisation of Al is same as B in $Na(BH_4)$.
 (B) Geometry around Al is same as in $AlCl_4^-$
 (C) AlH_4^- , BH_4^- , $AlCl_4^-$, are iso structural
 (D) AlH_4^- , BH_4^- , $AlCl_4^-$, are iso electronic
108. Select correct statements about $Al(BH_4)_3$
- (A) All three tetrahydrido borate form two hydrogen bridges
 (B) two BH_4^- form two hydrogen bridges and one form one hydrogen bridge
 (C) one BH_4^- form two hydrogen bridges and two form one hydrogen bridge
 (D) B form only 2c-2e bond



109. Total number of 2c–2e bond and 3c–2e bond in $\text{Al}(\text{BH}_4)_3$ are respectively
 (A) 6, 12 (B) 6, 6 (C) 12, 12 (D) 12, 6
110. Total number of 2c – 2e bond and 3c – 2e bond in $\text{Be}(\text{BH}_4)_2$ are respectively
 (A) 8, 4 (B) 4, 8 (C) 4, 4 (D) 8, 8

PART - IV : PRACTICE TEST-2 (IIT-JEE (ADVANCED Pattern))

Max. Time : 1 Hr.

Max. Marks : 66

Important Instructions

A. General :

- The test is of 1 hour duration.
- The Test Booklet consists of 22 questions. The maximum marks are 66.

B. Question Paper Format

- Each part consists of five sections.
- Section-1 contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE is correct.
- Section-2 contains 5 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE OR MORE THAN ONE are correct.
- Section-3 contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9 (both inclusive).
- Section-4 contains 1 paragraphs each describing theory, experiment and data etc. 3 questions relate to paragraph. Each question pertaining to a particular passage should have only one correct answer among the four given choices (A), (B), (C) and (D).
- Section-5 contains 1 multiple choice questions. Question has two lists (list-1 : P, Q, R and S; List-2 : 1, 2, 3 and 4). The options for the correct match are provided as (A), (B), (C) and (D) out of which ONLY ONE is correct.

C. Marking Scheme

- For each question in Section 1, 4 and 5 you will be awarded 3 marks if you darken the bubble corresponding to the correct answer and zero mark if no bubble is darkened. In all other cases, minus one (– 1) mark will be awarded.
- For each question in Section 2, you will be awarded 3 marks. If you darken all the bubble(s) corresponding to the correct answer(s) and zero mark. If no bubbles are darkened. No negative marks will be answered for incorrect answer in this section.
- For each question in Section 3, you will be awarded 3 marks if you darken only the bubble corresponding to the correct answer and zero mark if no bubble is darkened. No negative marks will be awarded for incorrect answer in this section.

SECTION-1 : (Only One option correct Type)

This section contains 7 multiple choice questions. Each questions has four choices (A), (B), (C) and (D) out of which Only ONE option is correct.

- Select the correct order of following property
 (A) % s-character : $\text{sp}^3 > \text{sp}^2 > \text{sp}$
 (B) O–N–O bond angle : $\text{NO}_3^- > \text{NO}_2^+ > \text{NO}_2$
 (C) All angles in CH_2F_2 are not identical
 (D) C–F bond length : $\text{CF}_4 > \text{CH}_3\text{F} > \text{CH}_2\text{F}_2 > \text{CF}_3\text{H}$
- F–As–F bond angle in AsF_3Cl_2 can be nearly :
 (A) 90° & 180° only (B) 120° only (C) 90° & 120° only (D) 90° only
- The correct sequence of polarity of the following molecule :
 1. Benzene 2. Inorganic Benzene 3. PCl_3F_2 4. PCl_2F_3
 (P stands for polar and NP stands for non-polar)

	1	2	2	4		1	2	2	4
(A)	P	NP	NP	P	(B)	NP	NP	NP	P
(C)	NP	P	NP	P	(D)	NP	P	P	NP



4. Which of the following molecule(s) is/are having $p\pi - p\pi$ back bonding
(A) BF_3 (B) BeF_2 (C) BCl_3 (D) All of these
5. In which of following option the (X–A–X) adjacent angle difference in cationic part and anionic part is maximum in the solid state.
(A) PCl_5 (B) Cl_2O_6 (C) N_2O_5 (D) BeH_2
6. The boiling point of a compound is raised by -
(A) intermolecular hydrogen bonding (B) High volatility
(C) Intramolecular hydrogen bonding (D) Non-polarity
7. In which of the following compound peroxide linkage is absent ?
(A) Per oxy mono sulphuric acid (B) Per titanate acid
(C) Per chloric acid (D) Per nitric acid

Section-2 : (One or More than one options correct Type)

This section contains 5 multipole choice questions. Each questions has four choices (A), (B), (C) and (D) out of which ONE or MORE THAN ONE are correct.

8. The molecule is/are having N–N bond.
(A) N_2O (B) N_2O_3 (C) N_2O_5 (D) N_2O_4
9. London force works in
(A) Polar molecule (B) Non-polar molecule
(C) All polar and non-polar molecule (D) Only in polar molecule
10. There is change in the type of hybridisation when :
(A) NH_3 combines with H^+ (B) AlH_3 combines with H^-
(C) NH_3 forms NH_2^- (D) SiF_4 forms SiF_6^{2-}
11. Select the correct order -
(A) Bond length = $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
(B) Bond strength = $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
(C) Unpaired electron(s) = $\text{O}_2 > \text{O}_2^+ > \text{O}_2^- > \text{O}_2^{2-}$
(D) No. of antibonding electron(s) = $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
12. Choose the option(s) regarding correct order of acidic nature :
(A) $\text{MgO} < \text{ZnO} < \text{P}_2\text{O}_5 < \text{SO}_3$ (B) $\text{MgO} < \text{ZnO} < \text{SO}_3 < \text{P}_2\text{O}_5$
(C) $\text{Li}_2\text{O} < \text{NO} < \text{CO}_2 < \text{SO}_2$ (D) $\text{Li}_2\text{O} < \text{BeO} < \text{CO}_2 < \text{NO}$

Section-3 : (One Integer Value Correct Type.)

This section contains 6 questions. Each question, when worked out will result in one integer from 0 to 9 (both inclusive)

13. X is the number of maximum atom(s) is/are present in same plane of $\text{B}_3\text{N}_3\text{H}_6$. Find value of X / 2
14. Find the number of molecules which are not having 3C–2e bond from the following :
(a) $\text{Al}_2(\text{CH}_3)_6$ (b) Si_2H_6 (c) B_2H_6 (d) C_2H_6
(e) Si_2Cl_6 (f) Al_2Cl_6 (g) $\text{B}_3\text{N}_3\text{H}_6$
15. Which of the following species having axial bond length greater than equatorial bond length :
 PCl_5 , IF_7 , SF_6 , CCl_4
16. The total number of orbitals involved in the overlapping in XeO_2F_4 ? Give answer after dividing by 4.
17. How many different type of bonds can be formed if any two d orbitals of two different atoms are overlapping with each other ?
18. In which of the following all bond length are not equal ?
(a) PCl_5 (b) SF_4 (c) ClF_3 (d) XeF_2
(e) $[\text{SF}_5]^+$ (f) $[\text{ClF}_4]^+$ (g) $[\text{XeF}_3]^+$ (h) O_3 (i) P_4 (white)

**SECTION-4 : Comprehension Type (Only One options correct)**

This section contains 1 paragraphs, each describing theory, experiments, data etc. 3 questions relate to the paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D)

Paragraph for Questions 19 to 21

Hybridisation is the mixing of atomic orbital of comparable energy and the number of hybrid orbitals formed is equal to the number of pure atomic orbitals mixed up and hybrid orbitals are occupied by σ -bond pair and lone pair.

19. Which of the following geometry is most unlikely to form from sp^3d hybridisation of the central atom
 (A) Linear (B) Regular Tetrahedral
 (C) T-shaped (D) See-Saw
20. The orbital, which does not participate in sp^3d^2 hybridisation.
 (A) p_x (B) d_{xy} (C) $d_{x^2-y^2}$ (D) p_z
21. "The hybrid orbitals are at angle of X° to one another" this statement is not valid for which of the following hybridisation. ($X \leq 180^\circ$)
 (A) sp^3 (B) sp^2 (C) sp^3d^2 (D) sp

SECTION-5 : Matching List Type (Only One options correct)

This section contains 1 questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D) out of which one is correct

22. Match each List-I with List-II and select the correct answer using the code given below the lists.

	List-I		List-II
	(Pair of species)		(Identical Property in pairs of species)
(P)	PCl_3F_2, PCl_2F_3	(1)	Hybridisation of central atom
(Q)	BF_3 & BCl_3	(2)	Shape of molecule / ion
(R)	CO_2 & CN_2^{-2}	(3)	μ (dipole moment)
(S)	C_6H_6 & $B_3N_3H_6$	(4)	Total number of electrons

Code :

	P	Q	R	S
(A)	1,2	1,2,3	1,2,3,4	1,2,3,4
(B)	1,2,3,4	1,2,3	1,2	1,2,3,4
(C)	1,2,3	1,2	1,2,3,4	1,2,3,4
(D)	1,2,3,4	1,2,3	1,2,3,4	1,2

Practice Test-2 ((IIT-JEE (ADVANCED Pattern))**OBJECTIVE RESPONSE SHEET (ORS)**

Que.	1	2	3	4	5	6	7	8	9	10
Ans.										
Que.	11	12	13	14	15	16	17	18	19	20
Ans.										
Que.	21	22								
Ans.										





APSP Answers

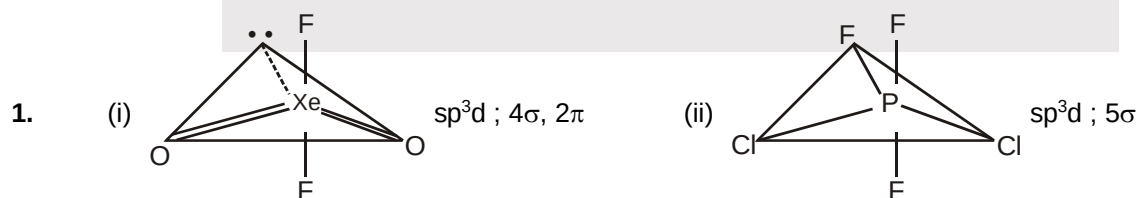
PART - I

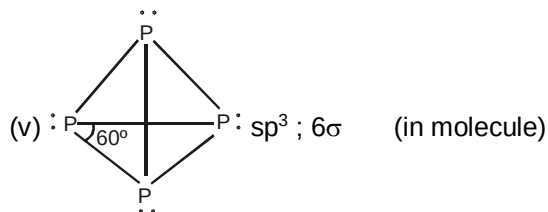
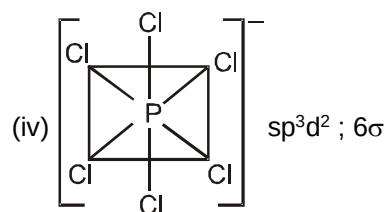
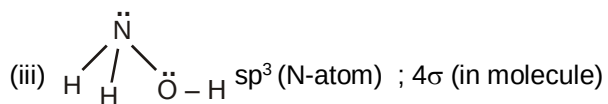
1.	(4)	2.	(4)	3.	(4)	4.	(3)	5.	(2)
6.	(3)	7.	(4)	8.	(3)	9.	(3)	10.	(2)
11.	(1)	12.	(2)	13.	(3)	14.	(1)	15.	(1)
16.	(1)	17.	(2)	18.	(2)	19.	(3)	20.	(2)
21.	(2)	22.	(4)	23.	(1)	24.	(4)	25.	(4)
26.	(2)	27.	(2)	28.	(2)	29.	(3)	30.	(2)

PART - II

1.	(B)	2.	(C)	3.	(A)	4.	(B)	5.	(C)
6.	(A)	7.	(C)	8.	(A)	9.	(C)	10.	(B)
11.	(D)	12.	(D)	13.	(D)	14.	(A)	15.	(D)
16.	(B)	17.	(C)	18.	(A)	19.	(B)	20.	(C)
21.	(C)	22.	(A)	23.	(D)	24.	(D)	25.	(D)
26.	(A)	27.	(D)	28.	(A)	29.	(C)	30.*	(CD)
31.	(C)	32.	(A)	33.	(D)	34.	(B)	35.	(B)
36.	(C)	37.	(D)	38.	(A)	39.	(C)	40.	(B)
41.	(B)	42.	(D)	43.	(C)	44.	(D)	45.	(B)
46.	(C)	47.	(C)	48.	(A)	49.	(D)	50.	(C)
51.	(C)	52.	(B)	53.	(B)	54.	(A)	55.	(D)
56.	(C)	57.	(A)	58.	(D)	59.	(C)	60.	(C)
61.*	(BC)	62.	(C)	63.	(B)	64.	(C)	65.	(B)
66.	(B)	67.	(D)	68.	(B)	69.	(D)	70.	(B)
71.	(D)	72.	(C)	73.	(A)	74.	(A)	75.	(A)
76.	(C)	77.	(D)	78.*	(A or C)	79.	(C)	80.	(B)
81.	(D)	82.	(D)						

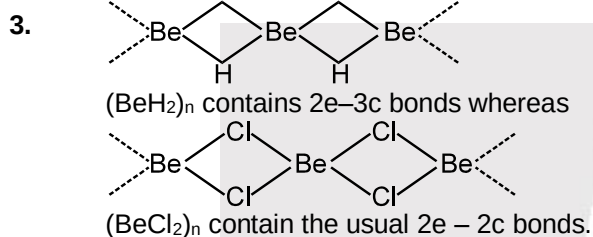
PART - III



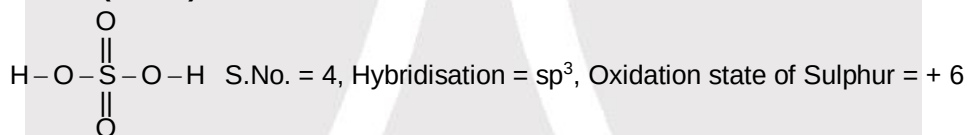


2.

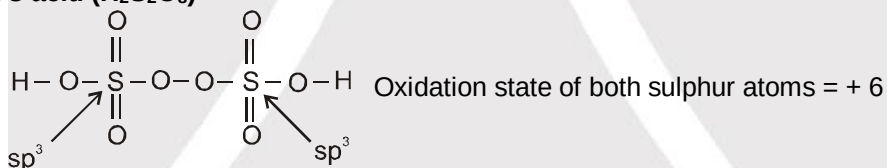
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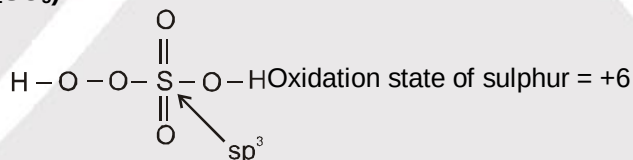
4. **Sulphuric acid (H_2SO_4)**



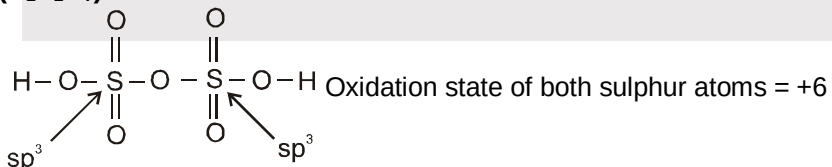
Marshall's acid ($H_2S_2O_8$)



Caro's acid (H_2SO_5)



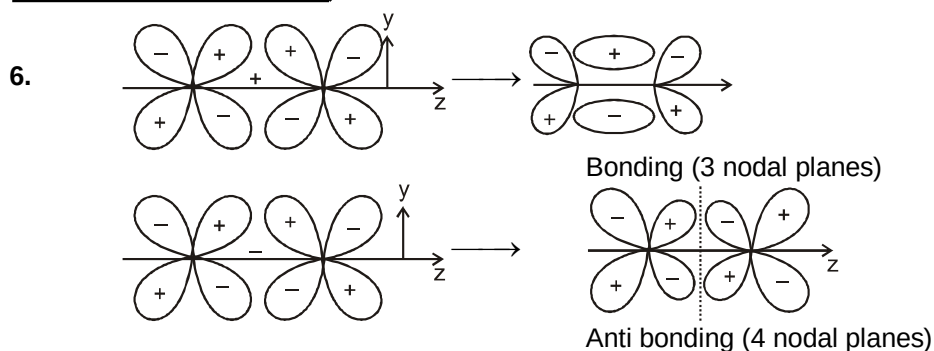
Oleum ($H_2S_2O_7$)



5.

3	1	2	4
P	Q	R	S





7. (i) $\text{Sc}^{2+} < \text{Ti}^{3+} < \text{V}^{3+} < \text{Cr}^{3+}$

(ii) $\text{Cd}^{2+} < \text{Zn}^{2+} < \text{Hg}^{2+}$

8. It has highest polarising power due to smallest ionic radius amongst alkali metal, therefore, greater degree of hydration is observed in Li^+ salts.

- | | | | | |
|--|---|---|------------|-------------|
| 9. (D) | 10. (C) | 11. (B) | 12. (C) | 13. (C) |
| 14. (D) | 15. (B) | 16. (C) | 17. (C) | 18. (B) |
| 19. (D) | 20. (A) | 21. (A) | 22. (B) | 23. (A) |
| 24. (C) | 25. (D) | 26. (D) | 27. (D) | 28. (A) |
| 29. (B) | 30. (B) | 31. (C) | 32. (B) | 33. (D) |
| 34. (C) | 35. (C) | 36. (D) | 37. (B) | 38. (A) |
| 39. (C) | 40. (B) | 41. (D) | 42. (D) | 43. (B) |
| 44. (C) | 45. (D) | 46. (B) | 47. (A) | 48. (C) |
| 49. (C) | 50. (B) | 51. (C) | 52. (D) | 53. (D) |
| 54. (B) | 55. (D) | 56. (C) | 57. (C) | 58. (B) |
| 59. (D) | 60. (A) | 61. (C) | 62. (A) | 63. (C) |
| 64. (A) | 65. (C) | 66. (A) | 67. (A) | 68. (B) |
| 69. (C) | 70. (A - p) ; (B - q) ; (C - p) ; (D - s) | 71. (A - r) ; (B - s) ; (C - p, s) ; (D - q, s) | | |
| 72. (A - p, r) ; (B - p, q) ; (C - p, r) ; (D - p, q, s) | | | | |
| 73. 13 (Except, i, iii, x, xiii, xviii) | 74. 6 | 75. 4 | 76. 75 | |
| 77. 25 | 78. 6 | 79. 4 | 80. 20 | 81. (ABCD) |
| 82. (BD) | 83. (BC) | 84. (BD) | 85. (AB) | 86. (AB) |
| 87. (AC) | 88. (AD) | 89. (BC) | 90. (ACD) | 91. (BCD) |
| 92. (BD) | 93. (BCD) | 94. (ABC) | 95. (ABCD) | 96. (ABCD) |
| 97. (ABC) | 98. (D) | 99. (ABC) | 100. (ABD) | 101. (ABCD) |
| 102. (ABD) | 103. (A) | 104. (D) | 105. (D) | 106. (B) |
| 107. (D) | 108. (A) | 109. (B) | 110. (C) | |

PART - IV

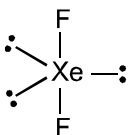
- | | | | | |
|-----------|----------|--------------------------|-----------------------|----------|
| 1. (C) | 2. (A) | 3. (B) | 4. (D) | 5. (C) |
| 6. (A) | 7. (C) | 8. (ABD) | 9. (ABC) | 10. (BD) |
| 11. (ABD) | 12. (AC) | 13. 6 | 14. 5 (b, d, e, f, g) | 15. 1 |
| 16. 4 | 17. 3 | 18. 6 (a, b, c, e, f, g) | 19. (B) | 20. (B) |
| 21. (C) | 22. (A) | | | |

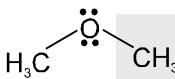


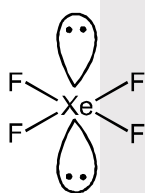
APSP Solutions

PART - I

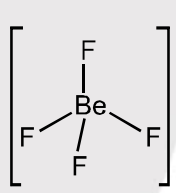
- Element belongs to d-block in d-block elements ($n - 1$) d and ns electron take part in the bonding.
- Proper overlap occurs between p_x and s orbital.
- $$\text{H}_3\text{N:} + \text{BF}_3 \longrightarrow \text{H}_3\text{N} \rightarrow \text{BF}_3$$

lewis base
lewis acid
- Bond energy $\propto$ Bond order.
- 

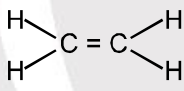
Steric number = 5 ; Hybridisation = sp^3d .
- 

Hybridisation = sp^3 ; Bond angle = 110°
- Boiling point of covalent molecules is decided by van der Waal force of attraction and van der Waal force of attraction depends on molecular weight.
- 

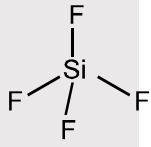
square planar
(All Xe-F bonds are equal)

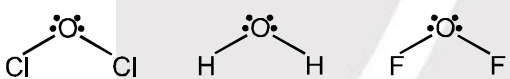


tetrahedral
(All Be-F bonds are equal)

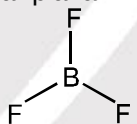


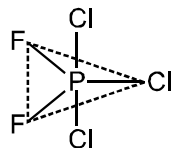
[C=C bond is different from C-H bond]



tetrahedral
(All Si-F bonds are equal)
- 

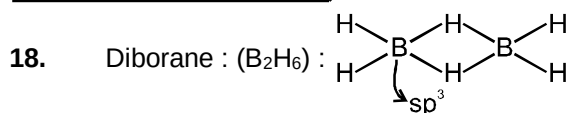
Cl-O-Cl bond angle is more due to large size of Cl and F-O-F bond angle is least due to bent rule.
- O_2^- has one unpaired electron in π^*2p_y while others do not have unpaired electrons. (explained by MOT)
- BF_3 has trigonal planar with no lone pair.



($\mu = 0$)
- Strongest H-bonding is in F-H O as F is most electronegative element hence creating good partial positive charge on H. While O is good donor of electron as compared to F due to its electronegativity.
- It is fact.
- Oxygen has unpaired in π^*2p_x and π^*2p_y which explain its paramagnetic behaviour. (Explained by MOT)
- 

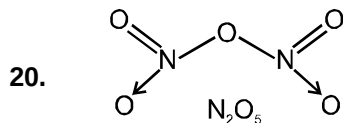
Trigonal bipyramidal
 $\mu = 0$



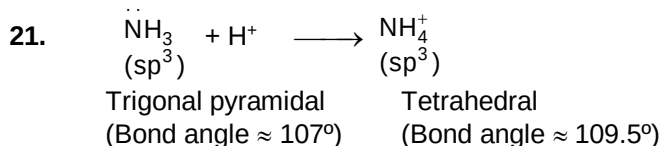


Diamond has sp^3 hybrid carbon and graphite has sp^2 hybrid carbon.

19. NH_4^+ , H_3O^+ and OF_2 all have sp^3 hybridisation and their shapes are tetrahedral, trigonal pyramidal and bent respectively.



It has both covalent as well as coordinate bond.



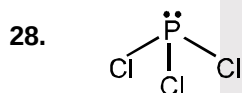
22. Polarisability of anion increases on increasing its size.

23. In CCl_4 , C has more charge therefore Cl is more polarized in CCl_4 .

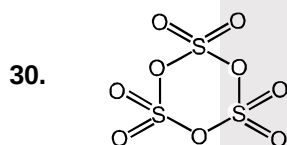
24. For the formation of ionic compound atom should have low ionization energy to form cation while other should have high electron affinity to form anion.

26. Covalent character in ionic compound is explained by Fajan's rule. Covalent character increases on decreasing the size of cation and increasing charge on cation.

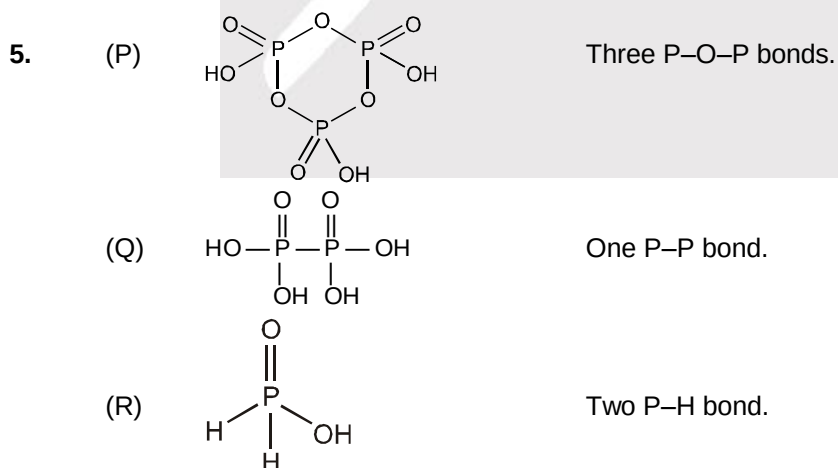
27. More is the electronegativity difference, stronger is the bond.

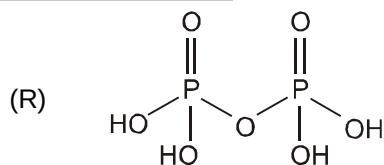


29. $O_2 \rightarrow [KK] \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \pi^* 2p_y^1$
 $O_2^- \rightarrow [KK] \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2 \pi^* 2p_y^1$
 As in O_2 HOMO is $\pi^* 2p_x$ and $\pi^* 2p_y$ and they have one electron each so next electron can be added to these orbitals.



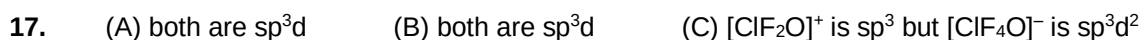
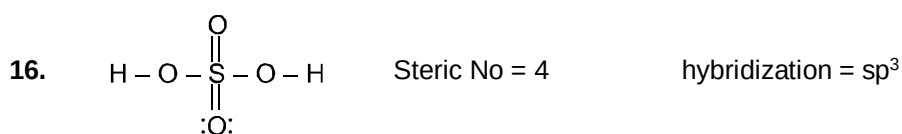
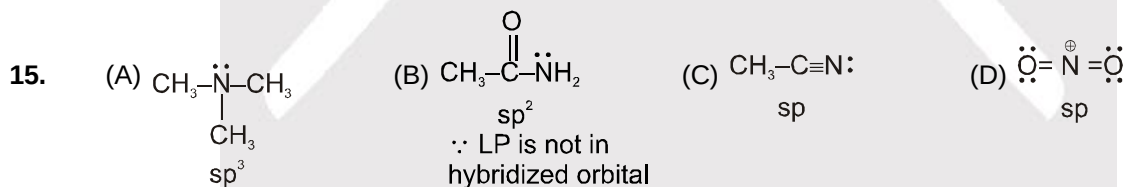
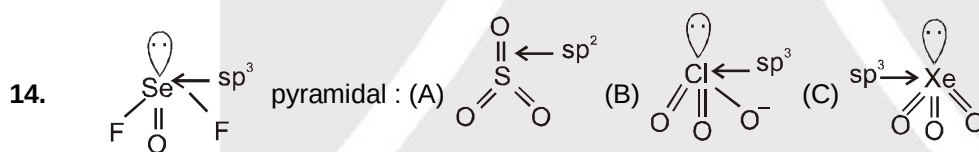
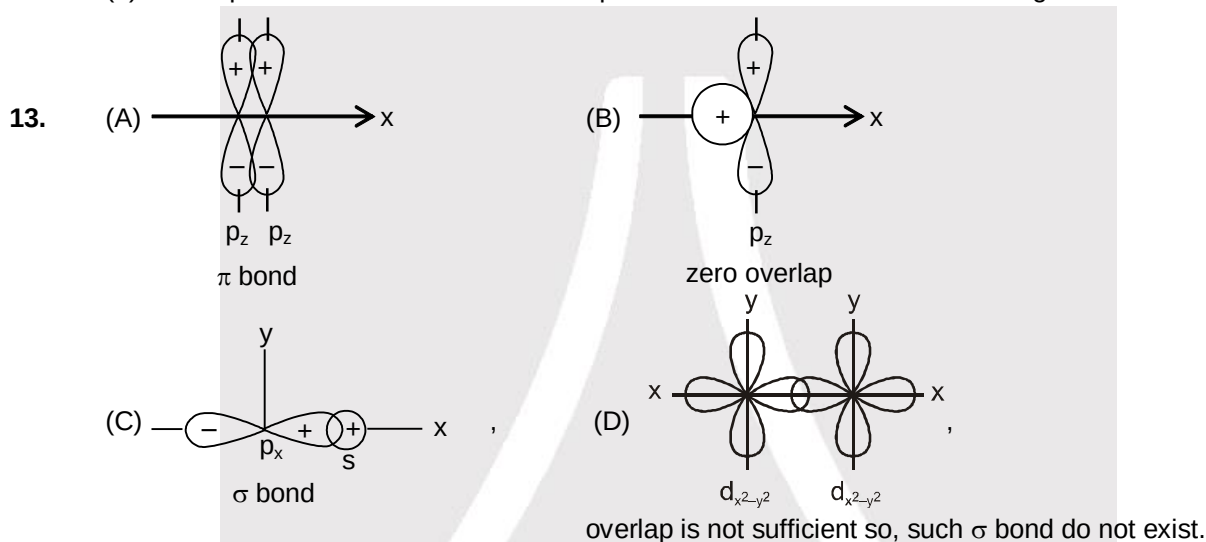
PART - III

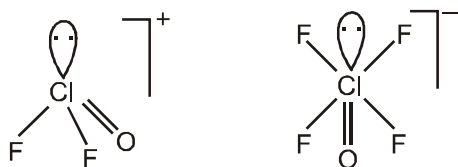




Four P–OH bond.

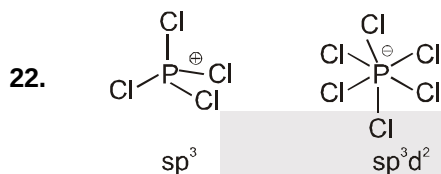
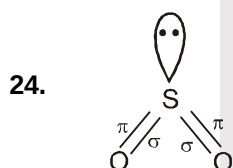
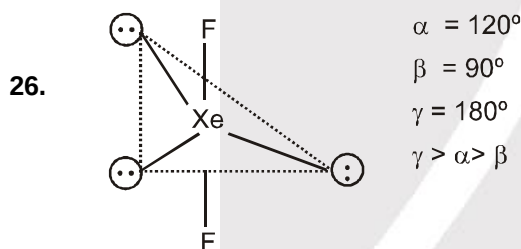
9. In SF_6 , PCl_5 and IF_7 the valence shell has 12, 10 and 14 electrons. As all contain more than 8 electrons in their valence shell they are example of super octet molecules.
10. Bond energy $\propto$ Bond order
- | Species | Bond order |
|--------------------|------------|
| CO | 3 |
| CO_2 | 2 |
| CO_3^{2-} | 1.33 |
11. Across a period, effective nuclear charge increases. As a result the size of atom, and, therefore, the size of d-orbital decreases leading to progressively stronger $p\pi-d\pi$ bonds.
12. (d) Position of atoms are different.
(e) Has unpaired electrons and this is not possible since the molecule is diamagnetic.



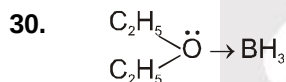
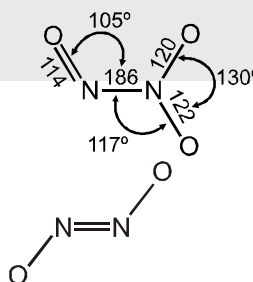
(D) both are sp^3d^2 18. The shape of AsF_4^- is see-saw.19. PH_3 – No Hybridisation NH_3 and H_2O bond angle is less than $109^\circ 28'$ due to LP–BP and LP–LP repulsion. CH_4 bond angle is $109^\circ 28'$

20. (i) Negative charge is more stable on 'S' than on N (size factor is dominant over EN factor).

(ii) Less formal charge provides more stability.

21. S–S overlap will result into σ bond.23. I and III are sp^3 hybridised while II and IV are sp^3 hybridised.25. P_4 bond angle = 60° PH_3 bond angle $\approx 90^\circ$ (Drago's rule) H_2O bond angle = 104.5° 

27. (A) Bent rule.

(C) H_2O_2 has more lone pair-lone pair repulsion.31. (A) N_2O_3 Dinitrogen trioxide(B) $\text{N}_2\text{O}_2^{2-}$ Hyponitrite ion

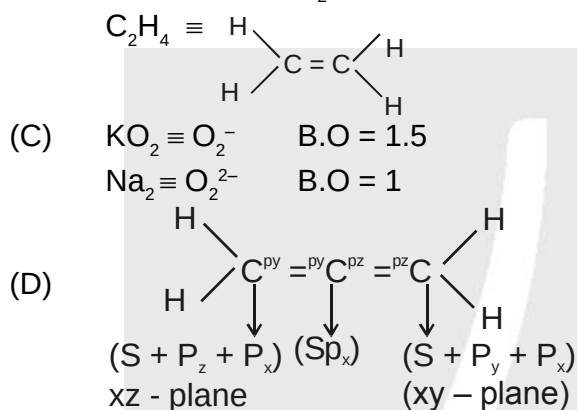


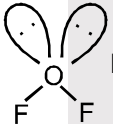
(C)	Bond Order	Internuclear Distance (pm)	Number of unpaired(s) Electrons
O_2^+ (dioxygenyl)	2.5	112.3	1
O_2 (dioxygen)	2.0	120.07	2
O_2^- (superoxide)	1.5	128	1
O_2^{2-} (peroxide)	1.0	149	0

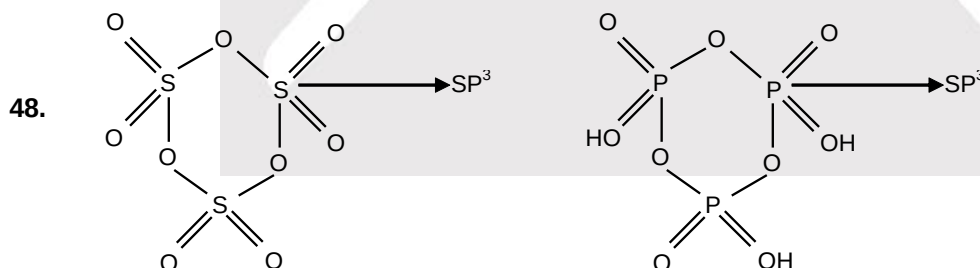
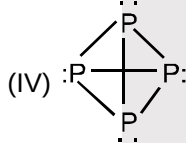
43. OF is derivative of O_2 and isoelectronic with O_2^- .
 So $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2p_z)^2 (\pi 2p_x^2 = \pi 2p_y^2) (\pi^* 2p_x^2 = \pi^* 2p_y^2)$
 The bond order of OF $1/2(10 - 7) = 1.5$.

44. ClO_2 has are unpaired electron.

45. (A) NO B.O = 3 triple bond.
 $O = N - Cl$ there is a double bond between N and 'O' atom
 (B) $CaC_2 \equiv Ca^{2+} + C_2^{2-} (-C \equiv C-)$

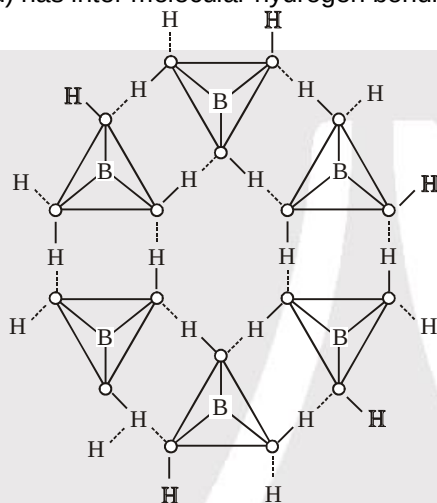


46. (I) $[PCl_4]^+ \rightarrow sp^3$
 (II)  has $\approx 102^\circ$ bond angle due to lp – lp repulsion, as bond pairs are closer to F-atoms.
 (III) All have sp^3 hybridisation and one lone pair.



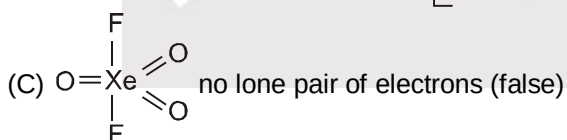
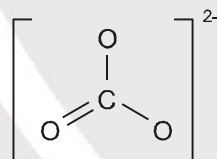


50. Dipole moment of compound if it would have been completely ionic
 $= (4.8 \times 10^{-10} \text{ esu}) (2.67 \times 10^{-8} \text{ cm}) = 12.8 \text{ D}$
 so % ionic character $= \frac{4.0}{12.8} \times 100\% = 31.25\%$
51. Graphite has layered structure. Layers are held by van der Waal's forces and distance between two layers is 340 pm. Each layer is composed of planar hexagonal rings of carbon atoms. C–C bond length within the layer is 141.5 pm. Each carbon atom in hexagonal ring undergoes sp^2 hybridisation and make three sigma bonds with three neighbouring carbon atoms. Fourth electron forms a π bond. The electrons are delocalised over the whole sheet.
53. (A) A fact
 (B) In the solid state and in liquid HF, the HF_2^- ion is held together by hydrogen bonding. In aqueous solutions there is hydrogen bonding but each HF molecule forms hydrogen bonds with the much more prevalent H_2O present, instead of with other HF molecules and H_3O^+ and F^- are much more likely to be formed.
 (C) H_3BO_3 (solid) has inter molecular hydrogen bonding.



54. (A) intermolecular hydrogen bonded. (true)

(B) structure of anions are different CO_3^{2-} – trigonal planar (sp^2)

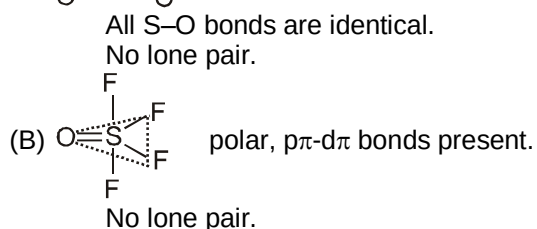
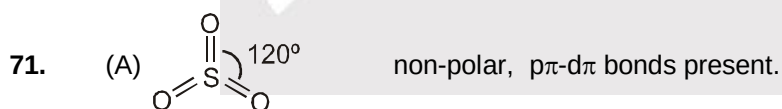


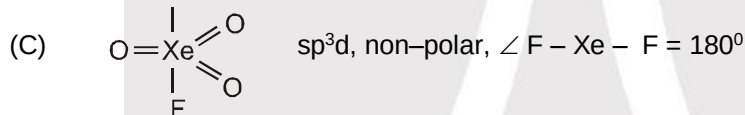
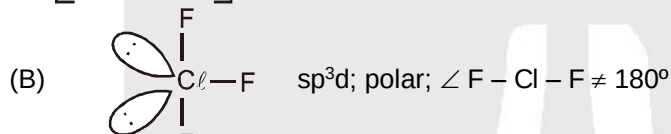
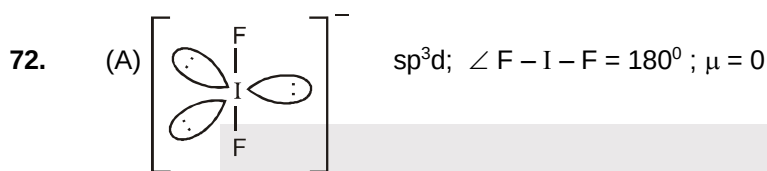
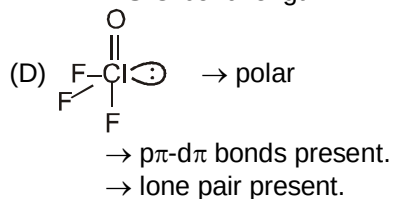
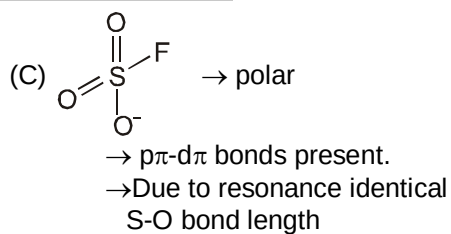
(D) True. $\text{D}_2\text{O} = 374.4 \text{ K}$; $\text{H}_2\text{O} = 373.0 \text{ K}$

55. D is less electronegative than H-atom so the results
56. This is based on the decreasing electronegativity difference between H(2.1) and Sb(1.9), As(2.0), P(2.1) and N(3.0).

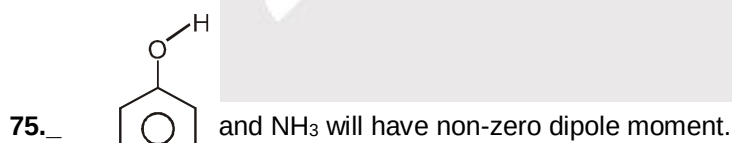
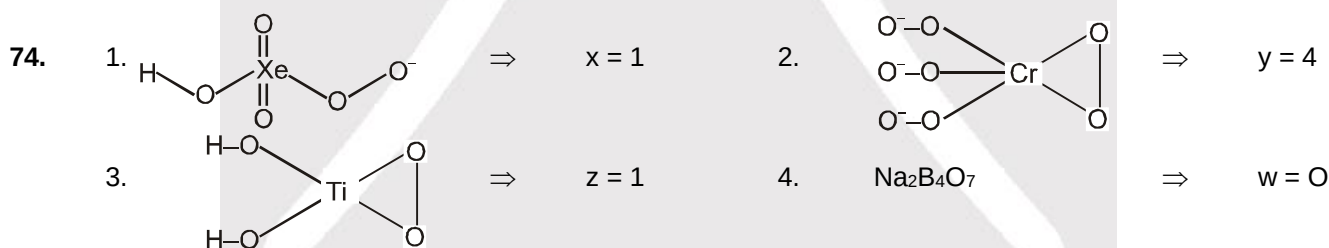


58. Fluorine is more electronegative but dipole moment is the product of charge and distance between combining atoms (covalently bonded) ; due to more charge separation in CH_3Cl , it has higher dipole moment.
59. (A) As δ^+ charge on central atom increases, the attraction between δ^+ and δ^- increases and thus $\text{Xe}-\text{F}$ bond length decreases. The correct order is $\text{XeF}_2 > \text{XeF}_4 > \text{XeF}_6$
 (B) PH_5 can not undergo sp^3d hybridisation as there is much large difference in size of s, p and d orbitals. PH_5 does not exist as no partial positive charge develops on P atom.
 (C) Dipole moment of CH_3Cl is greater than CH_3F due to greater charge separation on carbon and chlorine atoms in CH_3Cl .
 (D) it is a correct order.
 The strength of hydrogen bond depends upon :
 (i) size (ii) electronegativity and (iii) ease of donation of electron pair by electronegative element.
 Higher the value of electronegativity and smaller the size of the covalently bonded atom to H atom stronger is the hydrogen bonding.
60. Fact based
61. Fact based
62. NaCl is an ionic compound.
63. The element present in short period will be of 13th group (group no = $10 + 3 = 13$) i.e. aluminium and its oxide will be Al_2O_3 which is amphoteric in nature; as react with acids as well as bases forming salts and water.
64. As Pb^{2+} has low polarising power. So PbCl_2 is ionic.
65. Solubility BaCO_3 and MgCO_3 can be explained on the basis of their hydration and lattice energies.
66. AgI is less soluble than AgF due to covalent nature. LiI is more soluble than LiF due to high size difference of ions more ionic nature, more melting point. Higher the charge density, larger will be the hydrated radii.
67. Strength of metallic bond $\propto$ number of unpaired electrons.
68. (i) Alkali metal, potassium forms K^+ (number of electrons = 18) which is isoelectronic with P^{3-} (number of electrons = 18) $\rightarrow \text{Q}$.
 (ii) Transition element, chromium show maximum oxidation state of +6 in $\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} and they are coloured $\rightarrow \text{S}$.
 (iii) Noble gas, Kr has largest atomic radius because radius is expressed as vander Waal's radius and there is interelectronic repulsions due to completely filled outer most shell. Inert gas has highest first ionisation energy in the respective period because of stable valence shell electron configuration $\rightarrow \text{T}$.
 (iv) Element ($Z = 13$), aluminium has intermediate value of electronegativity and therefore its oxide is amphoteric in nature $\rightarrow \text{R}$.
70. Steric no. of $\text{I}_3^- = 5$, sp^3d ; Steric no. of $\text{XeO}_3 = 4$, sp^3 ;
 Steric no. of $\text{ClOF}_3 = 4 + 1 = 5$, sp^3d ; Steric no. of $\text{XeF}_5^+ = 5 + 1 = 6$, sp^3d^2





73. H_3PO_4 , H_3PO_3 , H_3PO_2 , $HClO_4$, $HClO_3$, $HClO_2$, H_2SO_3 , H_2SO_4 , H_2SO_5 , $H_2S_2O_8$, SO_3 , SO_2 , P_4O_{10}

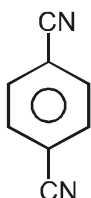


76. $\mu_{cal} = q \times d = 1.6 \times 10^{-19} \text{ C} \times 150 \times 10^{-12} \text{ m} = 2.4 \times 10^{-19} \text{ Cm}$

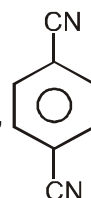
$\mu_{obs} = 1.8 \times 10^{-29} \text{ Cm}$

Percentage ionic character = $\frac{\mu_{cal}}{\mu_{obs}} \times 100 = 75\%$



77. $X = \text{BF}_3, \text{CO}_2, \text{PCl}_5, \text{CH}_4,$ 

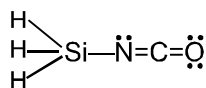
;

 $Y = \text{BF}_3, \text{CO}_2, \text{SO}_2, \text{ClF}_3,$ 

78. (1, 2, 5, 6, 7, 8)

79. (a) PCl_6^- (Coordinate Bond = 1)(b) $\text{NH}_3 \cdot \text{BF}_3$ (Coordinate Bond = 1)(c) HNO_3 (Coordinate Bond = 1)(D) CO (Coordinate Bond = 1)

80.



Vacant d-orbitals

LP of N can be delocalised back bonding ($p\pi - d\pi$). SN of N = 2 sp

81. The species in which the central atom has expanded octet, are called hypervalent compounds.

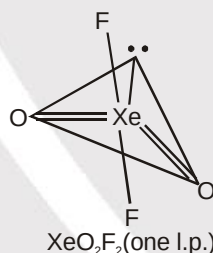
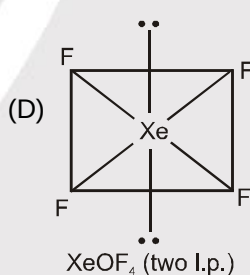
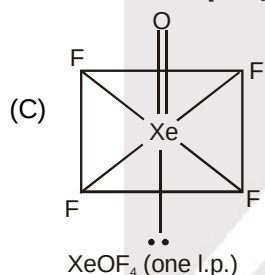
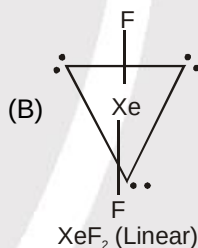
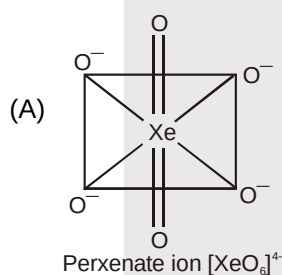
82. (b) If X-axis is internuclear axis, then it will result in π bond.(c) It will result in σ bond.

(d) & (e) It shows zero overlap with no bond formation.

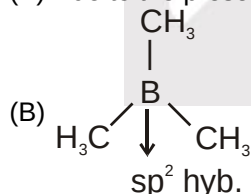
83. In PCl_6^- and BF_4^- , vacant orbital take part in hybridization

84. Bigger atoms, LP and double bond occupy equatorial positions in trigonal bipyramidal structure.

85.

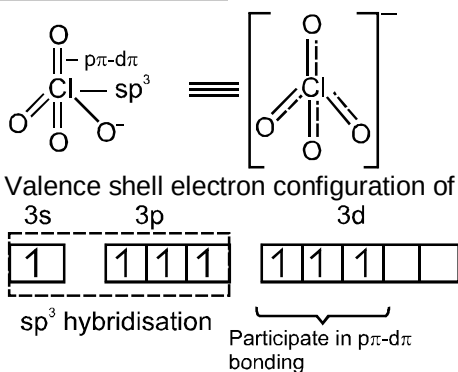


86. (A) Due to the presence of lone pair bond angle decreases.

(C) NH_4Cl is an ionic compound and 'N' is in sp^3 hybridisation.(D) S_8 molecule has 16 electron pairs left behind after the bonding.



87.



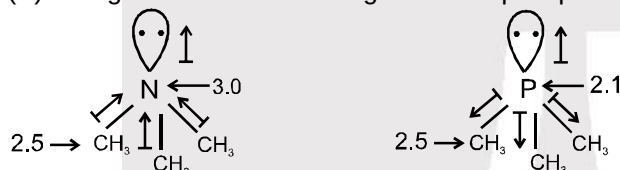
To obtain effective $p\pi-d\pi$ overlap the size of the d orbital must be similar to size of the p orbital. Hence in chlorine, $p\pi-d\pi$ bonding is so strong that no polymerization of oxoanions occurs.

89.

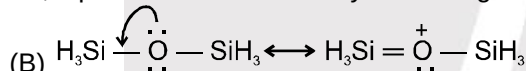
$(\text{CH}_3)_2\ddot{\text{O}}:$ can act as lewis base but $(\text{SiH}_3)_2\text{O}$ & $(\text{SiH}_3)_3\text{N}$ can not, as one pairs on O & N are delocalised in to empty orbitals of Si.

90.

(A) Nitrogen is more electronegative than phosphorus.



So, dipole moment of trimethylamine is greater than trimethyl phosphine.



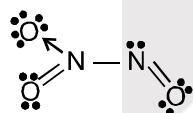
In trisilyl ether the lone pair of electron on oxygen atom is less easily available for donation because of $p\pi-d\pi$ delocalisation due to presence of the vacant d-orbital with Si. This however is not possible with carbon in $\text{CH}_3-\text{O}-\text{CH}_3$ due to the absence of d-orbital making it more basic.

(C) Bond order of C_2 and O_2 are same i.e., 2. In C_2 molecules both bonds are π -bonds whereas, there is one σ and one π -bond in O_2 molecule

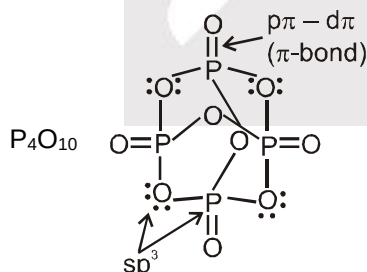
$\text{C}_2 = 131 \text{ pm}$; $\text{O}_2 = 121 \text{ pm}$.



91.



92.

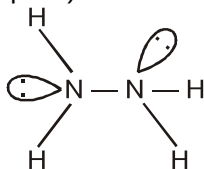


The P-O bond lengths shows that the bridging bonds on the edges are $1.60 \text{ \AA}$ but the P=O bonds on the corners are $1.43 \text{ \AA}$ and this P=O is formed by $p\pi-d\pi$ back bonding. A full p-orbital on the oxygen atom overlaps sideways with an empty d-orbital on the phosphorus atom. The bond angle POP is 127° and there is no P-P bonds.



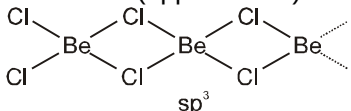


93. (A) Structure is similar to that of ethane. Each N atom is tetrahedrally surrounded by one N, two H and a lone pair. The two halves of the molecules are rotated 95° about N–N bond and occupy a gauche (non-eclipsed) conformation. The bond length is $1.45 \text{ \AA}$.

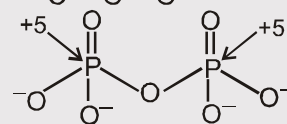
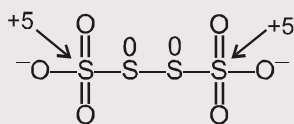
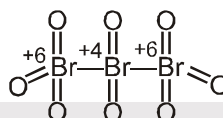
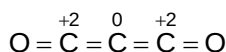


(B) Has partial double bond character due to $p\pi-d\pi$ delocalisation.

(C) $\text{OF}_2 = 103^\circ$ (approximate) and $\text{OCl}_2 = 112^\circ$ (approximate).

(D) Exist in polymeric structure as  in solid state.

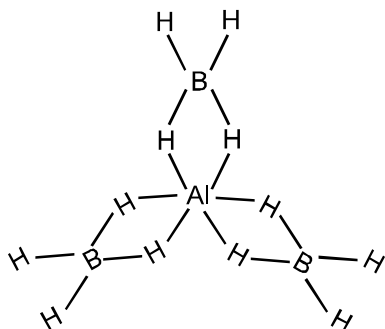
94. C_3O_2



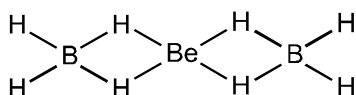
96. All the given species contain H-bonded to O, and so are capable of H-bonding.
98. The critical temperature depends on the magnitude (of strength) of intermolecular force of attraction between the molecules. If a molecule has dipole moment it means there is dipole-dipole attraction between the molecules and thus it will affect the critical temperature.
99. According to Fajan's rule.
100. Group 1 elements have lesser strength of metallic bonding than corresponding group 2 elements due to only one electron per atom participating in metallic bonding in group 1. In group 2 elements, two electrons per metal atom participate in metallic bonding.
101. (A) due to decreasing strength of metallic bond.
(B) Fact
(C) Elements having half-filled stable configurations have lesser strength of metallic bonding.
(D) Electron delocalisation is the main reason for electricity conduction in both.
102. (A) It results from the electrical attractions among positively charged metal ions and mobile, delocalised electrons belonging to the crystal as a whole.
(B) In general, greater the number of $(n-1)d$ and ns electrons, stronger is the resultant bonding.
(C) Strength of metallic bond depends on the type of hybrid orbitals participating in metallic bonding.
(D) As the size of atom increases the attraction between the positive part of the atom and delocalised electrons decreases and thus the strength of the metallic bond decreases.
103. A = Ne
104. B = P_4
105. $\text{BC}_3 = \text{PCl}_3$
 $\text{BC}_5 = \text{PCl}_5$



108.



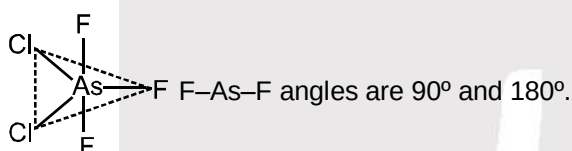
110.



PART - IV

1. In CH_2F_2 bond angle are not identical due to bent rule.

2.



3.

Benzene is non polar.
Inorganic benzene is non polar.



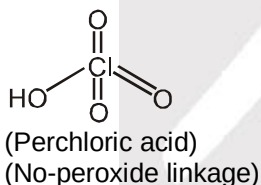
4.

In BF_3 , BeF_2 , BCl_3 $p\pi-p\pi$ back bonding takes place.
In BF_3 and BeF_2 $2p-2p\pi$ back bond and BCl_3 $2p-3p\pi$ back bond is formed.

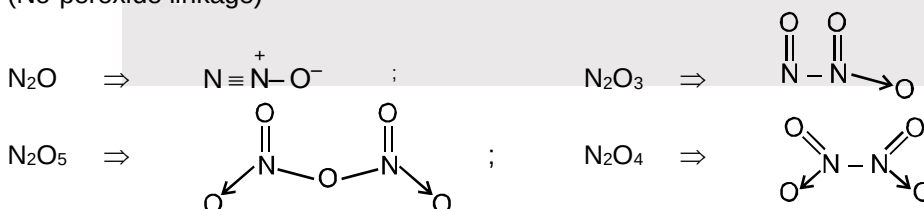
5.

$\text{PCl}_5 (\text{s}) \Rightarrow [\text{PCl}_4]^+ (109.5^\circ)$, $[\text{PCl}_6]^- (90^\circ)$
 $\text{Cl}_2\text{O}_6 (\text{s}) \Rightarrow [\text{ClO}_2]^+ (120^\circ)$, $[\text{ClO}_4]^- (109.5^\circ)$
 $\text{N}_2\text{O}_5 \Rightarrow [\text{NO}_2]^+ (180^\circ)$, $[\text{NO}_3]^- (120^\circ)$

7.



8.



9.

It is fact.

10.

$\text{NH}_3 (\text{sp}^3) + \text{H}^+ \longrightarrow \text{NH}_4^+ (\text{sp}^3)$; $\text{NH}_3 (\text{sp}^3) \longrightarrow \text{NH}_2^- (\text{sp}^3) + \text{H}^+$
 $\text{AlH}_3 (\text{sp}^2) + \text{H}^- \longrightarrow \text{AlH}_4^- (\text{sp}^3)$; $\text{SiF}_4 (\text{sp}^3) \longrightarrow \text{SiF}_6^{2-} (\text{sp}^3\text{d}^2)$

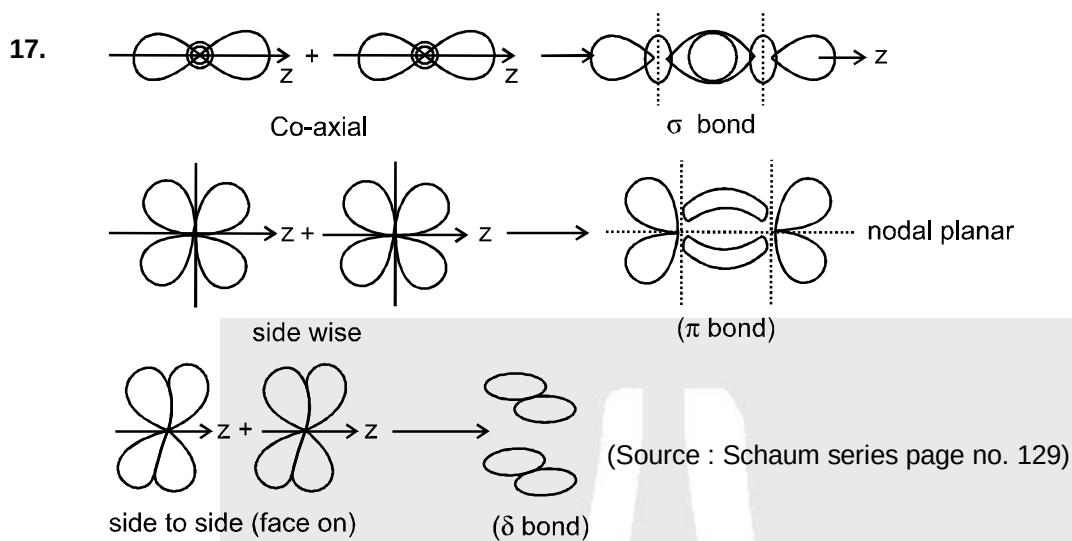




11. Using MOT, Bond order of $O_2^{2-} = 1$, $O_2^- = 1.5$, $O_2 = 2$, $O_2^+ = 2.5$.

15. PCl_5

16. Total no. of σ bonds = 6
 Total no. of π bonds = 2
 total no. of bonds = $6+2 = 8$
 No. of orbitals involved = $2 \times 8 = 16$



18. PCl_5 , SF_4 , ClF_3 , $[SF_5]^+$, $[ClF_4]^+$, $[XeF_3]^+$.
 $PCl_5 \rightarrow sp^3d$
 $SF_4 \rightarrow$
 $ClF_3 \rightarrow sp^3d$
 $[SF_5]^+ \rightarrow sp^3d$
 $[ClF_4]^+ \rightarrow sp^3d$
 $[XeF_3]^+ \rightarrow sp^3d$
 $[XeF_2] \rightarrow sp^3d$

